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INTERVIEW HANDBOOK SERIES

20

PAGES

OF HIGH-VALUE
INTERVIEW PREP

AWS

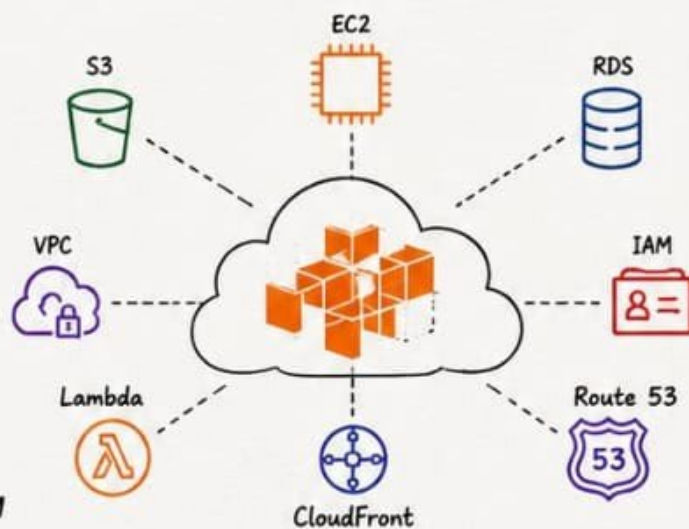
INTERVIEW HANDBOOK

Your Ultimate Guide to AWS Interview Success

✓ CORE SERVICES ✓ ARCHITECTURE ✓ TROUBLESHOOTING ✓ BEST PRACTICES

Interview
Ready!

- ✓ Top Questions
- ✓ Real Scenarios
- ✓ Architecture
- ✓ Deep Dives
- ✓ Troubleshooting
- ✓ Cheat Sheets



Think
Like an
Architect



Build.
Scale.
Succeed.



**COVER CORE
AWS SERVICES**

From fundamentals
to advanced topics



**MASTER
ARCHITECTURE**

Design scalable,
highly available
solutions



**TROUBLESHOOT
LIKE A PRO**

Identify issues,
analyze causes,
apply fixes



**SECURITY &
COMPLIANCE**

Understand IAM,
encryption, and
best practices



**CHEAT SHEETS &
QUICK REFERENCE**

Key facts, limits,
ports, and models
at a glance

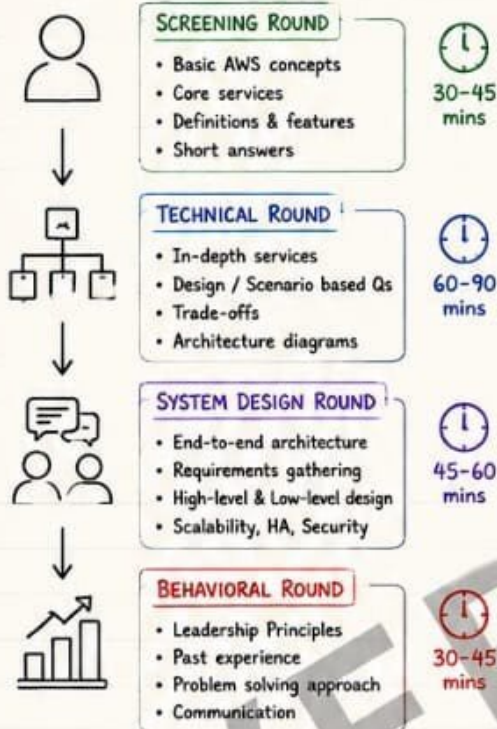
PREPARE SMART. INTERVIEW CONFIDENT. GET HIRED.

AWS INTERVIEW ROADMAP

"It's not about knowing every service.
It's about solving real problems with the right services."



1. AWS INTERVIEW STAGES



3. INTERVIEW FOCUS BY COMPANY TYPE

COMPANY TYPE	FOCUS AREAS	WHAT THEY CARE ABOUT
FAANG	- System Design - Scalability - Deep AWS Services - Performance	Can you design systems that scale to millions? Can you make the right trade-offs?
STARTUPS	- Cost Optimization - Serverless - Fast Iteration - Security Basics	Can you build fast? Can you save cost and ship reliable features?
CONSULTING (Firms)	- Wide AWS Knowledge - Best Practices - Well-Architected Framework - Client Communication	Can you apply AWS in different scenarios? Can you explain clearly and recommend?
ENTERPRISES	- Security & Compliance - Networking - HA / DR - Governance	Can you build secure, compliant and reliable systems at scale?

2. WHAT COMPANIES ACTUALLY TEST

- ✓ Core AWS Services & How they work together
- ✓ Understanding of Architecture & Trade-offs
- ✓ Scalability, High Availability, Fault Tolerance
- ✓ Security, IAM, Networking, Encryption
- ✓ Cost Optimization & Best Practices
- ✓ Monitoring, Logging & Troubleshooting
- ✓ Real-world Problem Solving
- ✓ Clear Communication & Structured Thinking



"Interviewers don't expect you to know everything.
They want to see how you think, design and make decisions."

4. AWS INTERVIEW TOPICS ROADMAP



5. WHAT YOU WILL GAIN FROM THIS HANDBOOK



Crack AWS Interviews with Confidence



Think like an Architect, not just an Engineer



Design Secure, Scalable and Highly Available Systems



Optimize Cost without compromising Performance



Real Scenarios, Real Questions, Real Answers



"The best way to predict your future in an interview is to prepare for it today."

— VERIQT

★ Master EC2 deeply. Many interviews start here.



1. WHAT IS AMAZON EC2?



Amazon EC2 (Elastic Compute Cloud) provides resizable compute capacity in the cloud. It allows you to run virtual servers (instances), configure security, storage, networking and scale as needed.

2. EC2 INSTANCE LIFECYCLE



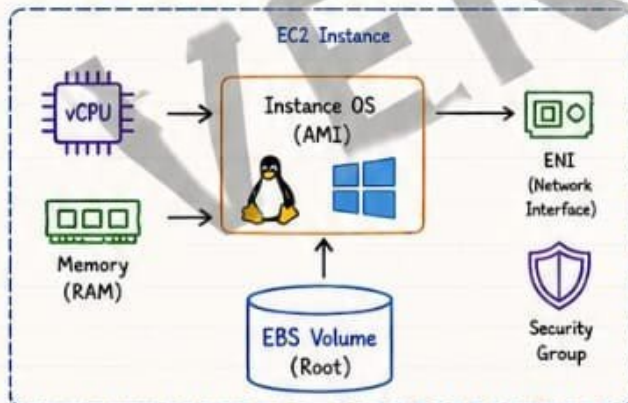
3. EC2 INSTANCE FAMILIES (COMMON)

FAMILY	USE CASE	EXAMPLES	CHARACTERISTICS
General Purpose	Balanced compute, memory, networking	t3, t3a, m5, m6i, t4g	Good for most applications
Compute Optimized	High performance compute	c5, c6i, c7g	Ideal for CPU intensive apps
Memory Optimized	In-memory databases, analytics	r5, r6i, x1e	High RAM instances
Storage Optimized	High I/O, large data processing	i3, i4i, d3	High disk IOPS and throughput
Accelerated Computing	GPU / ML / AI Workloads	p3, p4d, g5, trn1	GPU or AI accelerators

4. PRICING MODELS

On-Demand	Pay for compute by the second or hour with no commitment.
Reserved Instances	1 or 3 year commitment for significant discount (up to 72%).
Spot Instances	Unused EC2 capacity at up to 90% discount. Can be interrupted.
Savings Plans	Flexible pricing model for compute usage (EC2, Fargate, Lambda).
Dedicated Hosts	Physical server dedicated for your use.

5. EC2 INSTANCE COMPONENTS



★ EC2 instances are launched from AMIs and attached to EBS volumes.

6. COMMON INTERVIEW QUESTIONS

- Q. What is the difference between EC2 and ECS?
- A. EC2 provides virtual servers (infrastructure). ECS is a container orchestration service.
- Q. What is an AMI?
- A. Amazon Machine Image. A template used to launch EC2 instances.
- Q. What is the difference between Instance Store and EBS?
- A. Instance Store is ephemeral (lost on stop/terminate). EBS is network attached and persistent.
- Q. Can we change the instance type after launching?
- A. Yes, by stopping the instance (if EBS-backed) and then changing the instance type.
- Q. What happens if an AZ goes down?
- A. Instances in that AZ will be affected. Use Multi-AZ or Auto Scaling across multiple AZs.

7. EC2 STORAGE OPTIONS

EBS (Elastic Block Store)	Network attached block storage. Persistent and durable.
Instance Store	Local physical disks. Very fast but ephemeral.
EFS (Elastic File System)	Managed NFS file system. Shared across multiple instances.
S3	Object storage. Use for backups, static files, logs, archives.

8. EC2 BEST PRACTICES

- ✓ Use least privilege IAM roles for EC2.
- ✓ Store sensitive data in Secrets Manager or Parameter Store.
- ✓ Use Auto Scaling to handle traffic spikes.
- ✓ Place instances in private subnets. Use NAT Gateway for outbound.
- ✓ Enable detailed monitoring and CloudWatch alarms.
- ✓ Regularly patch and update your instances.
- ✓ Use EBS encryption and encrypted AMIs.



TIP: Always think in terms of scalability, availability, security and cost.

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VPC is the foundation of almost every AWS architecture. Understand it deeply!



Your isolated network in the AWS Cloud

1. WHAT IS AMAZON VPC?



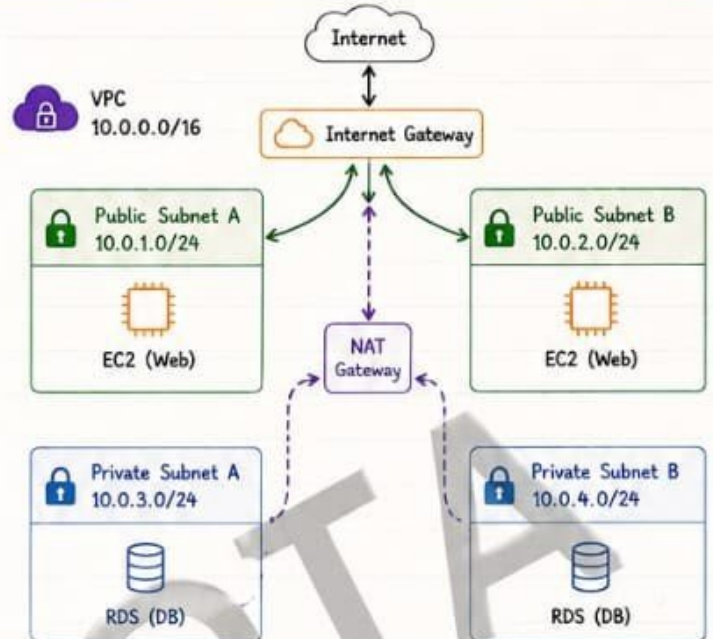
Amazon VPC (Virtual Private Cloud) lets you launch AWS resources in a virtual network that you define.

You have full control over your virtual networking environment.

2. VPC COMPONENTS

VPC	A logically isolated virtual network in the AWS Cloud.
Subnets	A range of IP addresses in your VPC. Can be public or private.
Route Table	Contains rules (routes) that determine where network traffic is directed.
Internet Gateway	Allows communication between your VPC and the internet.
NAT Gateway	Allows instances in private subnets to access the internet.
Security Group	Stateful firewall for your instances. Works at instance level.
Network ACL	Stateless firewall for subnets. Works at subnet level.
VPC Peering	Connects two VPCs privately.
Endpoints	Private connection to AWS services without internet gateway.

3. VPC NETWORKING OVERVIEW



Route Table (Example)	
Destination	Target
10.0.0.0/16	local
0.0.0.0/0	Internet Gateway
0.0.0.0/0 (Private Subnet)	NAT Gateway

→ Internet-bound traffic
--→ Private subnet internet access via NAT

4. PUBLIC VS PRIVATE SUBNET

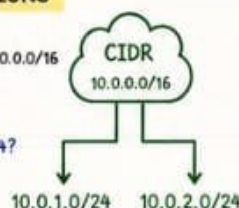
PUBLIC SUBNET	PRIVATE SUBNET
✓ Has route to Internet Gateway ✓ Instances have public IP ✓ Accessible from the internet ✓ Use for: ALB, Bastion, NAT GW	✓ No route to Internet Gateway ✓ Instances have no public IP ✓ Not accessible from internet ✓ Use for: App, DB, Backend
Diagram showing Internet connected to IGW, which connects to Public Subnet.	Diagram showing Internet connected to Private Subnet, but the connection is blocked (marked with a red X).
TIP: Keep your database in private subnets. Never expose DB to the internet.	

5. SECURITY GROUPS vs NACLs

SECURITY GROUP (SG)		NETWORK ACL (NACL)
Stateful	Type	Stateless
Works at Instance level	Scope	Works at Subnet level
Allow rules only	Rules	Allow and Deny rules
Default: Deny all inbound, Allow all outbound	Default	Default: Allow all inbound and outbound
Evaluated after NACL	Order	Evaluated before SG
Use case: Fine-grained control for instances	Best for	Use case: Subnet-level traffic control

6. COMMON CIDR & IP QUESTIONS

- Q: What is CIDR?
A: Classless Inter-Domain Routing. Example: 10.0.0.0/16
- Q: How many IPs in 10.0.0.0/24?
A: 256 IPs (254 usable)
- Q: How many subnets in 10.0.0.0/16 if /24?
A: 256 subnets
- Q: Can two subnets in a VPC overlap?
A: No. CIDR blocks must not overlap.



7. COMMON INTERVIEW QUESTIONS

- Q. What is the difference between Security Group and NACL?
- Q. How will you allow private instances to access internet?
- Q. How will you restrict SSH access to EC2?
- Q. How will you connect two VPCs?
- Q. How will you access an S3 bucket privately from your VPC?
- Q. What happens if you delete a route table?
- Q. Can two VPCs have the same CIDR?



KEY TAKEAWAY: VPC is all about isolation, control and connectivity. Master the components and traffic flow!

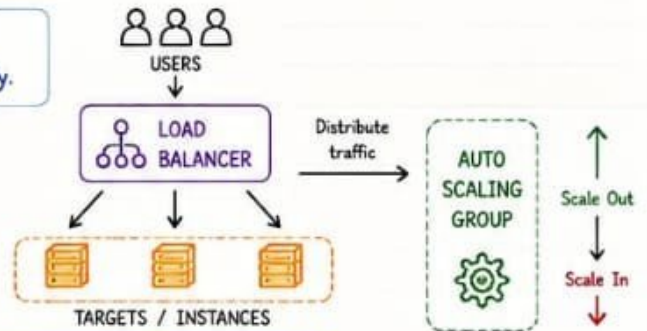
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Load Balancers distribute traffic across multiple targets.
Auto Scaling adjusts capacity to handle demand automatically.

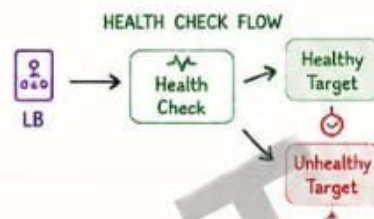
1. TYPES OF LOAD BALANCERS

Application Load Balancer (ALB)	- Layer 7 (HTTP/HTTPS) - Content-based routing - Host / Path based routing - Best for web applications
Network Load Balancer (NLB)	- Layer 4 (TCP/UDP) - Ultra high performance - Static IP & low latency - Best for high throughput
Classic Load Balancer (CLB)	- Legacy - Layer 4 & 7 - For existing applications - Supports EC2-Classic only



2. TARGET GROUPS & HEALTH CHECKS

- Target Group is a logical group of targets (instances, IPs, Lambda, etc.)
- Health checks ensure only healthy targets receive traffic.
- Unhealthy targets are removed automatically.



HEALTH CHECK SETTINGS

	Protocol: HTTP / HTTPS / TCP
	Port: Traffic Port / Custom
	Path: /health
	Healthy Threshold: 2 - 10
	Unhealthy Threshold: 2 - 10
	Interval: 5 - 300 sec
	Timeout: 2 - 120 sec

3. LOAD BALANCER COMPARISON

	ALB	NLB	CLB
Layer	Layer 7	Layer 4	Layer 4 & 7
Protocol	HTTP, HTTPS, gRPC, WebSocket	TCP, UDP, TLS	HTTP, HTTPS, TCP, SSL
Use Case	Web apps, Microservices	High performance, Gaming, IoT, Financial	Legacy applications
Routing	Host / Path based	IP / Port based	Limited
Performance	Good	Very High	Low
Static IP	No	Yes (Elastic IP)	No
WebSocket	Yes	Yes (TCP)	No
TLS Termination	Yes	Yes (TLS)	Yes
Cost	\$\$	\$\$	\$

4. AUTO SCALING GROUP (ASG)

- Ensures the right number of instances are running based on demand.
- Works with EC2 instances behind a Load Balancer.
- Replaces unhealthy instances automatically.

ASG COMPONENTS

- Launch Template / Configuration
- Min Size - Always On
- Desired Capacity - Target
- Max Size - Upper Limit
- Scaling Policies - Rules

5. AUTO SCALING POLICIES

Target Tracking Scaling

Maintains a target metric (e.g., CPU = 50%)

Step Scaling

Adds or removes instances based on metric thresholds

Simple Scaling

Adds or removes a fixed number of instances

Add / Remove fixed capacity

7. BEST PRACTICES

- Use ALB for web applications and NLB for high performance.
- Configure proper health checks. Don't send traffic to unhealthy instances.
- Use Target Tracking Scaling for most applications.
- Right-size your instances and use Scaling to optimize cost.
- Enable TLS termination at Load Balancer for security.
- Monitor with CloudWatch metrics and set alarms.



KEY TAKEAWAY: Use the right Load Balancer for the right use case and let Auto Scaling keep your application highly available and cost-efficient.

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6. COMMON INTERVIEW QUESTIONS

- Q. When would you choose ALB over NLB?
- A. When you need Layer 7 features like path based routing, host headers, or WebSocket support.
- Q. Can NLB route based on URL?
- A. No, NLB is Layer 4. It routes based on IP and Port.
- Q. What happens if all targets in a Target Group are unhealthy?
- A. The Load Balancer returns an error (503 for ALB, 502 for NLB) until targets become healthy.
- Q. How does Auto Scaling decide to add or remove instances?
- A. Based on CloudWatch metrics and scaling policies.
- Q. Difference between Desired Capacity and Max Size?
- A. Desired = target number of instances.
Max Size = upper limit ASG can reach.





IAM (Identity and Access Management) lets you securely control who can access AWS resources and what they can do.

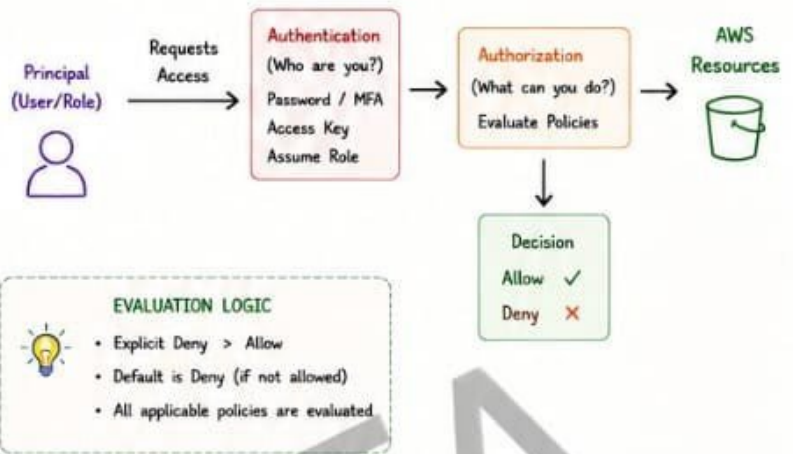


Secure access.
Least privilege.
Zero trust mindset.

1. IAM CORE COMPONENTS

COMPONENT	DESCRIPTION
Users	An identity with long-term credentials (Username + Password or Access Key).
Groups	A collection of users. Permissions are assigned to groups, not users.
Roles	An identity you can assume with temporary credentials. Used by AWS services, EC2, Lambda or external accounts.
Policies	JSON documents that define permissions. Attached to Users, Groups or Roles.
Permissions	Allow or Deny actions on AWS resources.
MFA	Adds an extra layer of security for human users.
Identity Providers	Manage federated access (SAML, OIDC) from external systems.

2. IAM ACCESS FLOW



3. POLICY TYPES

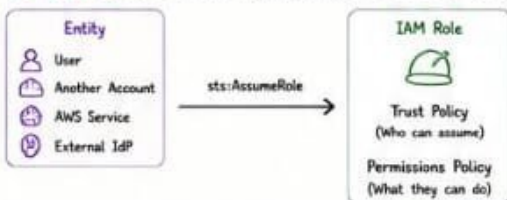
Identity-based Policy	Attached to Users, Groups, Roles. Controls what actions the identity can perform.
Resource-based Policy	Attached to a resource (S3 Bucket, Lambda, SNS, etc.) to control who can access that resource.
Permissions Boundary	Sets the maximum permissions an identity-based policy can grant.
Service Control Policy (SCP)	Applied at AWS Organizations level. Sets the maximum permissions for accounts in the organization.
Session Policy	Applied when assuming a role. Further restricts permissions in the session.

4. COMMON IAM INTERVIEW QUESTIONS

- Q. What is the difference between User and Role?
- A. A User represents a person or application with long-term credentials. A Role is assumed to get temporary credentials, often used by services.
- Q. What is the difference between Identity-based and Resource-based policy?
- A. Identity-based is attached to a user/role/group. Resource-based is attached to a resource and defines who can access it.
- Q. What is the difference between Allow and Deny?
- A. Deny always overrides Allow. If any policy denies, access is denied.
- Q. What is a Permissions Boundary?
- A. It sets the maximum permissions an entity can have, even if other policies allow more.
- Q. What is MFA and why is it important?
- A. MFA (Multi-Factor Authentication) adds an extra verification step to protect against credential theft.
- Q. How do you provide cross-account access?
- A. Use Roles with "sts:AssumeRole" permission and trust relationships.

5. TRUST RELATIONSHIP (FOR ROLES)

A trust policy (resource-based policy) defines who can assume the role.



Example Trust Policy (Allows EC2 to assume the role)

```

{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": { "Service": "ec2.amazonaws.com" },
      "Action": "sts:AssumeRole"
    }
  ]
}
  
```

6. COMMON IAM SCENARIOS

SCENARIO	SOLUTION
EC2 needs access to S3	Attach an IAM Role to EC2 with S3 permissions. Use instance profile.
User needs access to AWS Console	Create IAM User, add to Group, attach policy, enforce MFA.
Cross-account access (Dev to Prod)	Create Role in target account. Add trust relationship. Assume the role from source account.
Lambda needs access to DynamoDB	Attach IAM Role to Lambda with DynamoDB permissions.
Public read access to S3 bucket	Use Bucket Policy (Resource-based policy) to allow public read.

7. BEST PRACTICES



Apply least privilege access. Only give what is needed.



Use Groups to manage users at scale.



Enforce MFA for all human users.



Use Roles for applications and services.



Enable CloudTrail and monitor IAM activity.



Use Permission Boundaries and SCPs wisely.



Regularly review and rotate credentials.



KEY TAKEAWAY: Great IAM design = Least privilege + Strong authentication + Continuous monitoring.

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AWS S3 INTERVIEW GUIDE

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Amazon S3 is designed for 99.999999999% (11 9's) durability and 99.99% availability.

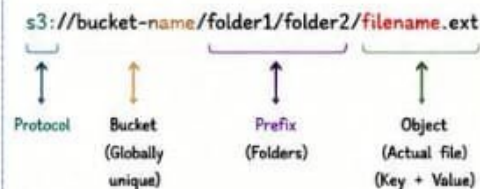


Simple Storage Service (S3) is object storage built to store and retrieve any amount of data from anywhere.

1. WHAT IS AMAZON S3?

- Object storage service that stores data as objects in buckets.
- Highly durable, scalable, secure and available.
- Use cases: Backup, Archive, Data lakes, Static websites, Media hosting, Logs, Big data analytics.

S3 OBJECT STRUCTURE



2. S3 STORAGE CLASSES (KEY DIFFERENCES)

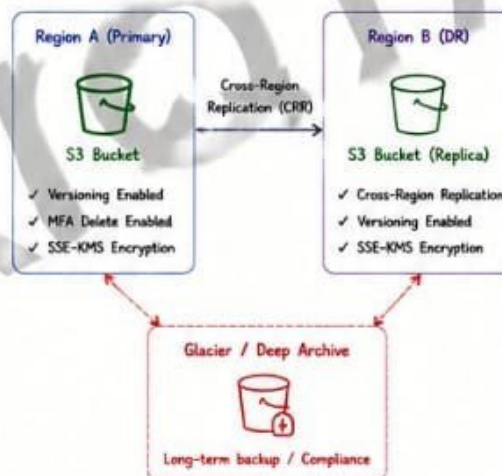
STORAGE CLASS	USE CASE	DURABILITY	AVAILABILITY	MIN DURATION	RETRIEVAL
Standard (General Purpose)	Frequently accessed data	11 9's	99.99%	None	Milliseconds
Intelligent-Tiering	Unknown or changing access patterns	11 9's	99.9%	None	Milliseconds
Standard-IA	Infrequently accessed data	11 9's	99.9%	30 days	Milliseconds
One Zone-IA	Re-creatable, infrequently accessed data	11 9's	99.5%	30 days	Milliseconds
Glacier Instant Retrieval	Archive data, instant access	11 9's	99.9%	90 days	Milliseconds
Glacier Flexible Retrieval	Archive data, flexible access	11 9's	99.9%	90 days	Minutes - Hours
Glacier Deep Archive	Long-term archive, lowest cost	11 9's	99.9%	180 days	Hours

★ Choose the right storage class to optimize performance and cost.

3. IMPORTANT S3 FEATURES

Versioning	Keeps multiple versions of an object in a bucket.
Lifecycle Policies	Automatically transition objects between storage classes or expire them.
Replication	Copy objects across regions (CRR) or within same region (SRR) for durability/latency.
Encryption	Protect data at rest using SSE-S3, SSE-KMS or SSE-C.
Bucket Policies	JSON based policies to control access to buckets and objects.
Object Lock	WORM (Write Once Read Many) for compliance. Prevent deletion or modification.
Static Website Hosting	Host static websites directly from S3.

4. S3 DATA PROTECTION



Best Practice:

- Enable Versioning
- Enable Encryption
- Use Lifecycle Policies
- Replicate Critical Data
- Monitor with CloudWatch

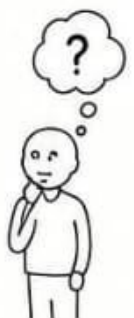
5. EXAMPLE: S3 BUCKET POLICY (PUBLIC READ)

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "PublicReadGetObject",
      "Effect": "Allow",
      "Principal": "*",
      "Action": "s3:GetObject",
      "Resource": "arn:aws:s3:::my-bucket/*"
    }
  ]
}
```

⚠ Never make sensitive buckets public!

6. COMMON INTERVIEW QUESTIONS

- Q. What is the difference between S3 and EBS?
A. S3 is object storage, used over the network. EBS is block storage, attached to EC2.
- Q. How do you reduce S3 storage cost?
A. Use appropriate storage class, lifecycle policies and delete unnecessary data.
- Q. How do you secure data in S3?
A. Use IAM policies, Bucket policies, Encryption and Block Public Access.
- Q. How do you share files with external users?
A. Use Pre-signed URLs or CloudFront with OAC/OAI.
- Q. Difference between Versioning and Lifecycle?
A. Versioning keeps multiple versions. Lifecycle moves or deletes objects automatically.



7. S3 COMMON USE CASES

Backup & Restore	Store backups, snapshots, and restore when needed.
Data Lakes	Store raw data for analytics and machine learning.
Static Websites	Host websites, portfolios, documentation.
Logs & Analytics	Store application logs, access logs, and analyze.
Media Hosting	Store images, videos, audio for global delivery.

8. S3 BEST PRACTICES

- | | |
|---|---|
| <ul style="list-style-type: none">✓ Enable Versioning on all critical buckets.✓ Enable Block Public Access.✓ Use SSE-KMS for encryption.✓ Use Lifecycle Policies to manage cost.✓ Replication for critical and compliance data. | <ul style="list-style-type: none">✓ Use IAM Roles (not access keys) for apps.✓ Monitor with CloudWatch metrics.✓ Enable Server Access Logging.✓ Use Pre-signed URLs for secure sharing.✓ Regularly review bucket permissions. |
|---|---|



KEY TAKEAWAY: S3 is all about durability, availability and cost optimization. Design for security, lifecycle and the right storage class.

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AWS RDS & AURORA

INTERVIEW GUIDE

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★ Amazon RDS makes it easy to set up, operate, and scale relational databases in the cloud. Aurora is AWS's high-performance, MySQL & PostgreSQL compatible database built for the cloud.

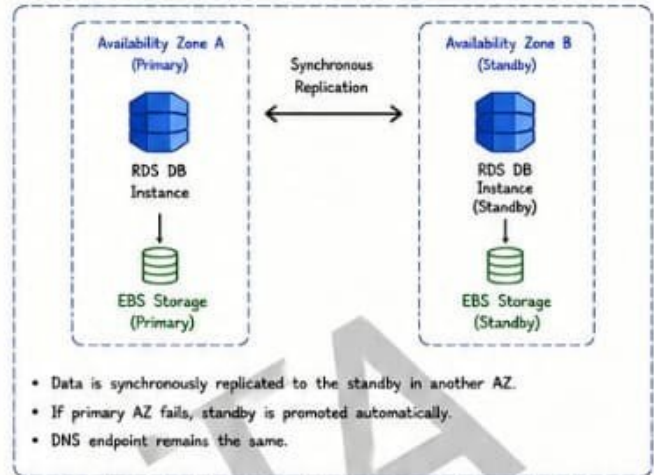


- ✓ Managed
- ✓ Scalable
- ✓ Secure
- ✓ Highly Available
- ✓ Cost-Effective

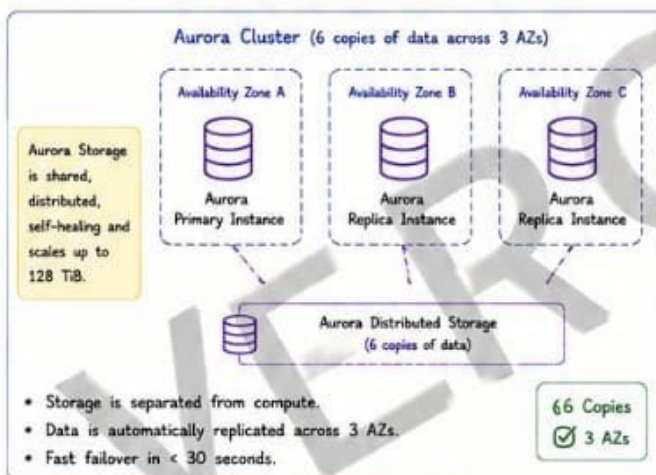
1. RDS vs AURORA – KEY DIFFERENCES

FEATURE	RDS	AURORA
Engines	MySQL, PostgreSQL, MariaDB, Oracle, SQL Server	Aurora MySQL, Aurora PostgreSQL
Performance	Good	Up to 5x MySQL, 3x PostgreSQL
Storage	EBS (General Purpose / IO) Limited to 64 TiB	Aurora Storage (Distributed) Up to 128 TiB
Availability	Multi-AZ	Multi-AZ by default (6 copies across 3 AZs)
Read Replicas	Yes	Yes (Up to 15)
Scaling	Vertical (Scale Up/Down)	Vertical + Serverless v2 (Auto scaling)
Failover Time	Minutes	< 30 seconds
Price	Pay for DB instance + storage	Pay for ACUs + storage
Best For	Traditional workloads	High performance, cloud-native workloads

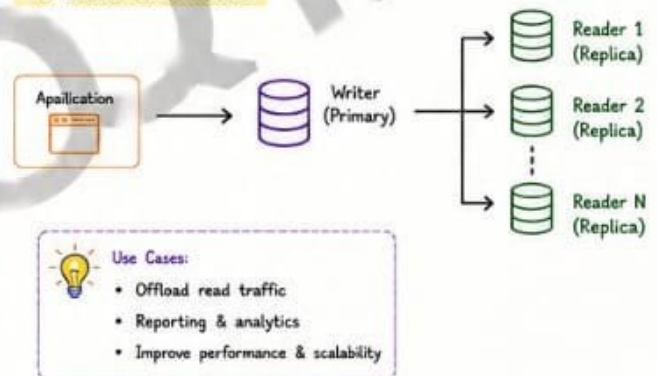
2. RDS ARCHITECTURE (MULTI-AZ)



3. AURORA ARCHITECTURE (HIGH AVAILABILITY)



4. READ REPLICAS



5. BACKUP & RECOVERY

Automated Backups	Enabled by default. Backups stored in S3. Retention: 1 - 35 days.
Manual Snapshots	Create on-demand snapshots and retain as long as needed.
Point-in-Time Recovery (PITR)	Restore your DB to any second within the retention period.
Cross-Region Snapshot Copy	Copy snapshots to another region for DR or compliance.
Aurora Backtrack	Backtrack your DB to a specific time (Up to 72 hours).

6. COMMON INTERVIEW QUESTIONS

- Q. What is the difference between RDS and Aurora?
- A. Aurora is AWS's own database engine with distributed storage, higher performance and built-in high availability.
- Q. How does Multi-AZ work in RDS?
- A. RDS synchronously replicates data to a standby in another AZ. If primary fails, standby is promoted.
- Q. How is Aurora different in High Availability?
- A. Aurora stores 6 copies of data across 3 AZs. Fails over in < 30 sec.
- Q. How do you scale an RDS database?
- A. Vertically by scaling the instance up or down. Read replicas for reads.
- Q. How do you scale Aurora?
- A. Vertically, horizontally (read replicas) and Serverless v2.



7. COMMON ENGINE PORTS

ENGINE	DEFAULT PORT
MySQL / Aurora MySQL	3306
PostgreSQL / Aurora PostgreSQL	5432
MariaDB	3306
Oracle	1521
SQL Server	1433



Ensure Security Groups allow only required ports from trusted sources.

8. BEST PRACTICES

- ✓ Use Multi-AZ for high availability.
- ✓ Enable automated backups and test restore regularly.
- ✓ Use Read Replicas for read-heavy workloads.
- ✓ Use IAM DB Authentication for secure connections.
- ✓ Encrypt data using KMS.
- ✓ Monitor using CloudWatch (CPU, Connections, Latency, Storage).
- ✓ Keep minor version updates up to date.



KEY TAKEAWAY:

Choose **RDS** for simplicity and compatibility.
Choose **Aurora** for performance, scalability and high availability.

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AWS ROUTE 53

INTERVIEW GUIDE

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- ★ Amazon Route 53 is a highly available and scalable DNS (Domain Name System) web service. It routes end users to internet applications by translating domain names into IP addresses.



- ✓ Highly Available
- ✓ Scalable
- ✓ Low Latency
- ✓ Domain Registration
- ✓ Health Check Integration

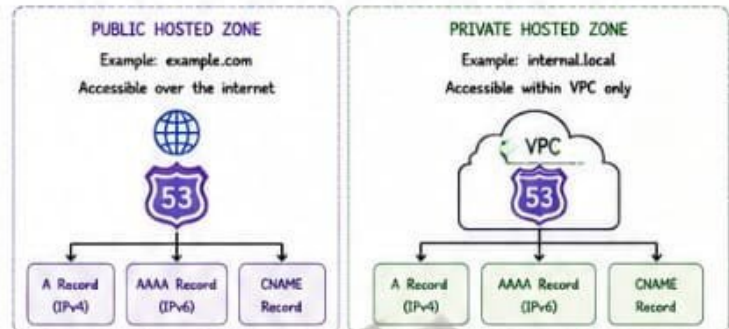
1. WHAT IS ROUTE 53?

- Managed DNS service from AWS.
- Converts domain names into IP addresses.
- Routes users to healthy resources.
- Works with AWS services & external resources.
- Supports routing policies for different use cases.



2. HOSTED ZONES

- A container for DNS records for a domain.
- Two types:
 1. Public Hosted Zone - Used for domains on the internet.
 2. Private Hosted Zone - Used within VPC(s) for internal domains.



3. ROUTING POLICIES

POLICY	DESCRIPTION	USE CASE
Simple Routing	Routes traffic to a single resource.	Basic use cases with one resource.
Weighted Routing	Routes traffic to multiple resources based on weights.	Load balancing, canary deployments.
Latency Routing	Routes traffic to the region with the lowest latency.	Improve performance for global users.
Failover Routing	Routes to a primary resource. If it fails, route to secondary.	Disaster recovery, high availability.
Geolocation Routing	Routes traffic based on the geographic location of users.	Serve content based on country or region.
Multi-Value Answer	Returns multiple healthy records to Route 53.	High availability, simple load balancing.
Geoproximity Routing	Routes traffic based on the location of users and resources.	Advanced use cases for global applications.

4. HEALTH CHECKS

- Monitors the health of your resources.
- Works with Route 53 to route traffic only to healthy resources.
- Supports:

- HTTP / HTTPS
- TCP
- CloudWatch Alarms
- Calculate Health Checks



TIP:
Always use Health Checks with Failover and Latency Routing.

5. DNS RECORD TYPES (COMMON)

RECORD	DESCRIPTION	EXAMPLE
A	Maps domain to IPv4 address	example.com → 1.2.3.4
AAAA	Maps domain to IPv6 address	example.com → 2001:db8::1
CNAME	Alias of one domain to another	www → example.com
MX	Mail exchange record	example.com → mail.example.com
NS	Name server record	example.com → ns-123.awsdns.com
TXT	Text record (SPF, verification, etc.)	example.com → "v=spf1 include:_spf.example.com ~all"
SRV	Service locator record	_sip._tcp.example.com → 10 60 5066 sip

6. COMMON INTERVIEW QUESTIONS

- Q. What is Route 53?
A. A highly available and scalable DNS web service by AWS.
- Q. What is the difference between public and private hosted zones?
A. Public is accessible over the internet, private is accessible within VPC.
- Q. Which routing policy is best for disaster recovery?
A. Failover Routing Policy.
- Q. How does Route 53 improve application performance?
A. By using Latency or Geolocation routing policies.
- Q. What is the purpose of health checks?
A. To monitor resource health and route traffic only to healthy resources.
- Q. Can Route 53 be used for non-AWS resources?
A. Yes, it can route traffic to any internet-accessible resource.



7. EXAMPLE: FAILOVER ROUTING



8. BEST PRACTICES

- ✓ Use meaningful and consistent domain naming.
- ✓ Enable DNSSEC for domain security.
- ✓ Use TTL carefully (low for changes, high for stability).
- ✓ Use health checks with Failover and Latency routing.
- ✓ Enable CloudWatch metrics for monitoring.
- ✓ Use Alias records for AWS resources (better performance & HA).



★ **KEY TAKEAWAY:** Route 53 is the backbone of DNS in AWS.

Master routing policies, health checks and best practices for reliable architectures.

- VERIQTA

AWS CLOUDFRONT INTERVIEW GUIDE

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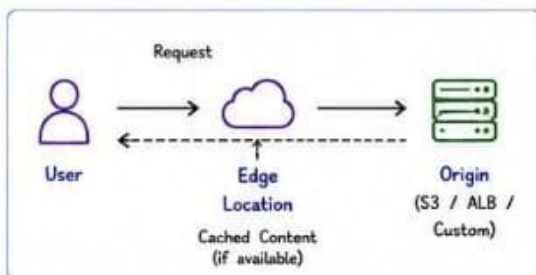
Amazon CloudFront is a fast content delivery network (CDN) service that securely delivers data, videos, applications, and APIs to customers globally with low latency and high transfer speeds.



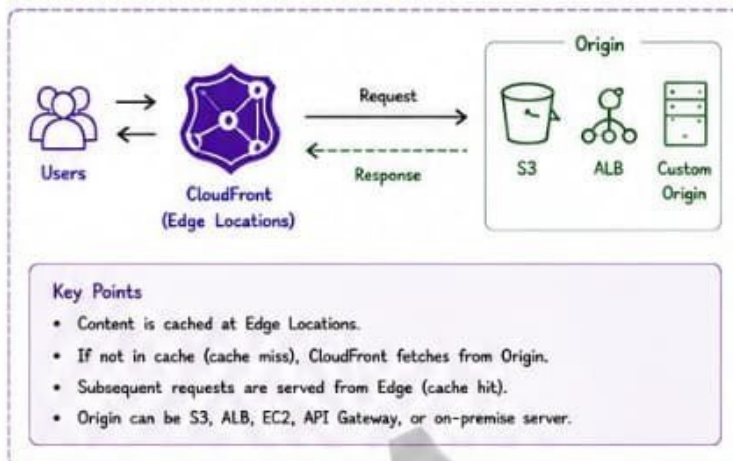
- ✓ Global CDN
- ✓ Low Latency
- ✓ Secure
- ✓ Scalable
- ✓ Cost-Effective

1. WHAT IS CLOUDFRONT?

- Global Content Delivery Network (CDN) from AWS.
- Delivers content to users with low latency and high transfer speeds.
- Content is cached at Edge Locations around the world.
- Integrates with other AWS services and custom origins.
- Uses Anycast IP and AWS Global Network.



2. CLOUDFRONT ARCHITECTURE



3. ORIGIN TYPES

ORIGIN TYPE	DESCRIPTION
S3 Origin	Best for static content (images, videos, docs, web assets). Supports OAC/OAI for secure access.
Custom Origin (ALB)	For dynamic content from Application Load Balancer.
Custom Origin (EC2)	For content from EC2 instance or any HTTP server.
API Gateway Origin	For APIs. Supports private APIs with authentication.
On-Premise / Other	Use CloudFront with Origin over the internet or VPN.

4. CACHING BEHAVIOR

CONCEPT	DETAILS
Cache Hit	Content found in Edge Location cache. Fast response.
Cache Miss	Content not in cache. Fetched from Origin.
TTL (Time to Live)	Controls how long content stays in cache. Min TTL, Default TTL, Max TTL.
Invalidation	Manually clear cached content before TTL expires.
Cache Key	Determined by path, headers, cookies, query strings.



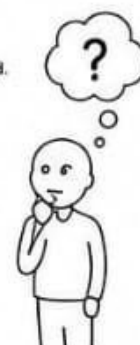
TIP: Use proper cache policies to improve hit ratio and reduce origin load.

5. SECURITY FEATURES

OAC (Origin Access Control)	Recommended way to securely access S3 origins using SigV4 authentication.
OAI (Origin Access Identity)	Legacy method to access S3 origins.
HTTPS	Encrypts data in transit using TLS/SSL.
Signed URLs	Grant access to private content using signed URLs with expiry.
Signed Cookies	Grant access to multiple files using signed cookies.
Geo Restriction	Allow or block content based on country.
AWS WAF Integration	Protect your content from common web exploits.

6. COMMON INTERVIEW QUESTIONS

- Q. What is CloudFront?
- A. A global CDN service that delivers content with low latency and high speed.
- Q. How does CloudFront improve performance?
- A. By caching content at Edge Locations closer to users.
- Q. What are the origin types supported by CloudFront?
- A. S3, ALB, EC2, API Gateway, and custom origins.
- Q. What is the difference between OAC and OAI?
- A. OAC uses SigV4 auth (recommended), OAI is legacy.
- Q. What is TTL in CloudFront?
- A. Time to Live controls how long content is cached at Edge.
- Q. How do you invalidate content in CloudFront?
- A. Use CloudFront Invalidation to remove cached objects.
- Q. Can CloudFront be used for dynamic content?
- A. Yes, using custom origins like ALB, EC2, or API Gateway.



7. CLOUDFRONT PRICING FACTORS

- Data Transfer Out (by region)
- Requests (per 10,000)
- Invalidation Requests
- Field-Level Encryption
- Real-Time Logs (Kinesis Data Streams)



8. BEST PRACTICES

Use OAC for S3 origins for better security.	Use appropriate TTLs to balance freshness and performance.
Compress content (Gzip/Brotli) to reduce transfer size.	Use cache policies instead of forwarding all headers/cookies.
Enable HTTP/2 or HTTP/3 for faster performance.	Monitor with CloudWatch metrics and Real-Time Logs.



KEY TAKEAWAY:

CloudFront brings your content closer to users, reduces latency, improves availability and offloads traffic from your origin. Design smart cache policies and secure your origins.

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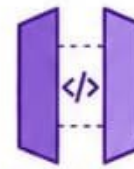
AWS API GATEWAY

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Amazon API Gateway is a fully managed service that makes it easy for developers to create, publish, maintain, monitor, and secure APIs at any scale.

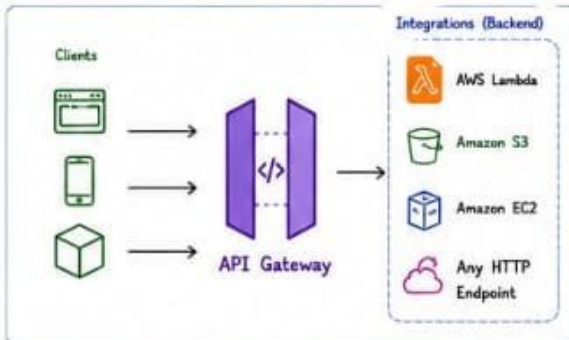


API GATEWAY

- ✓ Fully Managed
- ✓ Secure
- ✓ Scalable
- ✓ Highly Available
- ✓ Pay Per Request

1. WHAT IS API GATEWAY?

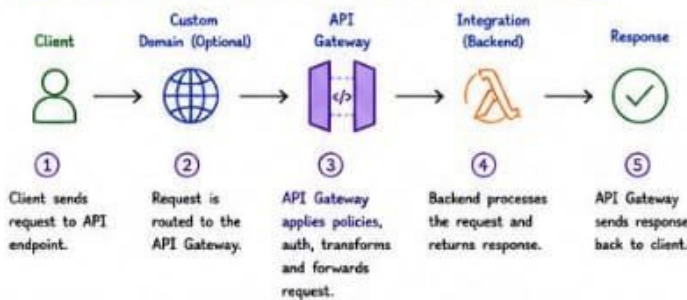
- Fully managed service to create, publish, and secure APIs.
- Acts as a front door for applications to access backend services.
- Handles authentication, authorization, throttling, caching, and monitoring.
- Works with AWS services and any HTTP backend.



2. API TYPES

TYPE	DESCRIPTION	USE CASES	KEY FEATURES
REST APIs	Build RESTful APIs that integrate with AWS services or any HTTP endpoint.	Mobile apps, Web apps, Microservices, Public APIs	• Resource & Method • Integrations • Request/Response Mapping • Caching, Throttling
HTTP APIs	Build low-latency HTTP APIs for multiple backends.	Real-time apps, Microservices, IoT backends	• Lower cost • Higher performance • JWT Authorizers • Built-in CORS
WebSocket APIs	Real-time, full-duplex communication between clients and backend.	Chat apps, Gaming, Live dashboards, Collaborative apps	• Persistent connections • Route-based messaging • Scale automatically
REST APIs (Private)	Expose REST APIs only within a VPC using AWS PrivateLink.	Internal applications, Enterprise systems	• Private access • Enhanced security • No public endpoint

3. API GATEWAY REQUEST FLOW (REST API EXAMPLE)



4. AUTHENTICATION & AUTHORIZATION

AUTH TYPE	DESCRIPTION	USE CASES
IAM	Use AWS IAM policies to control access.	Internal services, AWS-to-AWS calls
Lambda Authorizer	Custom auth logic using AWS Lambda.	Custom tokens, complex business logic
Cognito User Pools	Authenticate users using Amazon Cognito.	Mobile & Web apps with user sign-in
JWT Authorizer (HTTP APIs)	Validate JSON Web Tokens without Lambda.	High performance, low latency APIs
API Keys	Simple API key-based access control.	Partner APIs, basic access control

5. INTEGRATION TYPES

TYPE	DESCRIPTION
Lambda Function	Invoke AWS Lambda functions.
AWS Service	Integrate with AWS services (S3, DynamoDB, SQS, etc.).
HTTP	Forward request to any HTTP/HTTPS endpoint.
Mock	Generate a mock response without calling backend.
VPC Link	Access private resources in a VPC.

7. MONITORING & LOGGING

Amazon CloudWatch	Monitor metrics, latency, errors, usage, and logs.
Access Logs	Capture request details for debugging and auditing.
X-Ray Tracing	Trace requests end-to-end across services.
Detailed Metrics	Track 4XX/5XX errors, latency, throttles, cache hits, etc.

Key Metrics

- Count (Requests)
- Latency
- 4XX Errors / 5XX Errors
- Integration Latency
- Cache Hit Count
- Throttle Count

6. COMMON INTERVIEW QUESTIONS

- Q. What is API Gateway?
- A. A fully managed service to create, publish, secure, and monitor APIs.
- Q. Difference between REST APIs and HTTP APIs?
- A. HTTP APIs provide lower latency and cost, while REST APIs offer more features.
- Q. How does API Gateway handle authentication?
- A. Using IAM, Lambda Authorizer, Cognito, JWT, or API Keys.
- Q. What is a VPC Link?
- A. A connection to access resources in a VPC from API Gateway.
- Q. How do you reduce API latency?
- A. Use HTTP APIs, caching, optimize integrations, and reduce backend latency.
- Q. How do you control API usage?
- A. Use throttling, usage plans, and API keys.



8. BEST PRACTICES

- ✓ Use the right API type for your use case.
- ✓ Enable caching to reduce backend load and latency.
- ✓ Implement throttling and usage plans to control costs.
- ✓ Use Lambda Authorizers or JWT for secure access.
- ✓ Log and monitor APIs using CloudWatch and X-Ray.
- ✓ Use stage variables for configuration across environments.
- ✓ Version your APIs and use stages (dev, test, prod).
- ✓ Validate and sanitize inputs to prevent attacks.

Security Tips

- Use HTTPS only
- Enforce IAM or strong auth
- Validate inputs
- Use WAF for additional protection



KEY TAKEAWAY:

API Gateway simplifies building, securing and managing APIs.
Use the right API type, secure it properly, monitor everything, and optimize for performance.

— VERIQTA

AWS LAMBDA INTERVIEW GUIDE

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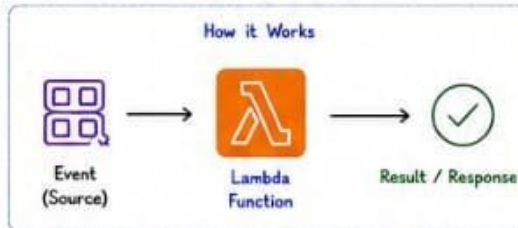


AWS Lambda is a compute service that lets you run code without provisioning or managing servers. You pay only for the compute time you consume.

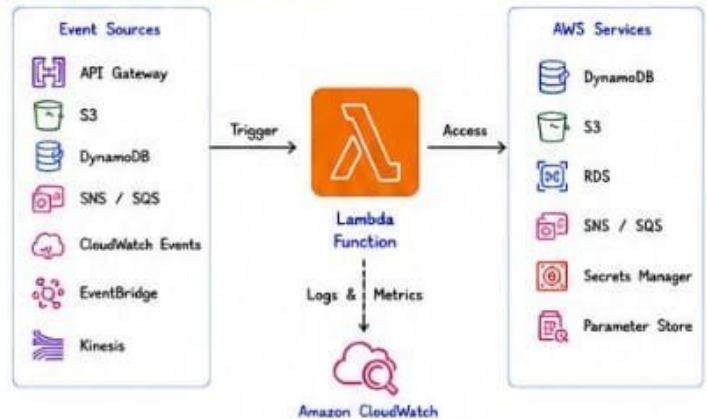
- ✓ Serverless
- ✓ Auto Scaling
- ✓ Pay Per Use
- ✓ Highly Available
- ✓ Integrated with AWS Services
- ✓ Event Driven

1. WHAT IS AWS LAMBDA?

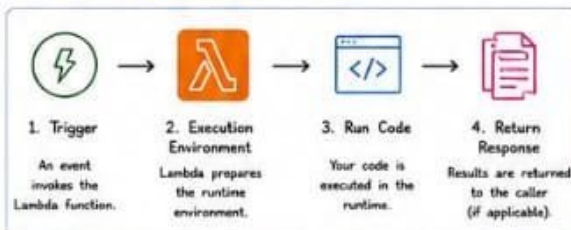
- Run code without provisioning or managing servers.
- Automatically scales based on the number of requests.
- Supports multiple programming languages.
- Integrated with many AWS services.
- Pay only for the duration of code execution.



2. LAMBDA ARCHITECTURE



3. LAMBDA EXECUTION MODEL



Key Points

- Lambda uses execution environments that are reused to reduce cold starts.
- If no environment is available, a new one is created (cold start).
- Each execution is isolated and secure.
- /tmp directory provides temporary storage (up to 10 GB).

4. LAMBDA FEATURES

FEATURE	DESCRIPTION
Serverless	No servers to manage. Lambda handles infrastructure.
Auto Scaling	Automatically scales up and down with demand.
Pay Per Use	Pay only for the compute time you consume.
Multiple Runtimes	Supports Python, Node.js, Java, .NET, Go, Ruby, and more.
High Availability	Built-in fault tolerance across multiple AZs.
Security	Fine-grained access control with IAM.
Integrations	Deep integration with AWS services.
Monitoring	Monitor with Amazon CloudWatch.

5. LAMBDA PRICING (PAY PER USE)

You pay only for the following:

- Requests: Number of requests.
- Duration: Total duration of execution (in GB-seconds).

$$\text{Price} = \text{Requests} \times \text{Request Price} + \text{Duration (GB-sec)} \times \text{Duration Price}$$



TIP: Optimize memory allocation and code to reduce duration and save costs.

7. COMMON USE CASES



8. LAMBDA LIMITS (DEFAULT)

LIMIT	DEFAULT VALUE
Maximum Execution Time	15 minutes
Memory Allocation	128 MB - 10,240 MB (10 GB)
/tmp Storage	Up to 10 GB
Deployment Package (ZIP)	50 MB (compressed)
Unzipped Code Size	250 MB
Environment Variables	4 KB
Concurrent Executions (Account Level)	1,000 (adjustable)

Supported Runtimes (Examples)

- Python
- Node.js
- Java
- .NET
- Go
- Ruby
- Custom Runtime

9. BEST PRACTICES

- ✓ Keep functions small and focused.
- ✓ Use appropriate memory allocation.
- ✓ Enable X-Ray for tracing and debugging.
- ✓ Use environment variables and Parameter Store for configs.
- ✓ Implement retries and handle exceptions properly.
- ✓ Store logs in CloudWatch and set up alarms.
- ✓ Use IAM roles with least privilege access.
- ✓ Use provisioned concurrency for critical workloads.



KEY TAKEAWAY:

AWS Lambda helps you run code without managing servers. Optimize for performance, cost, and observability.

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AWS SQS INTERVIEW GUIDE

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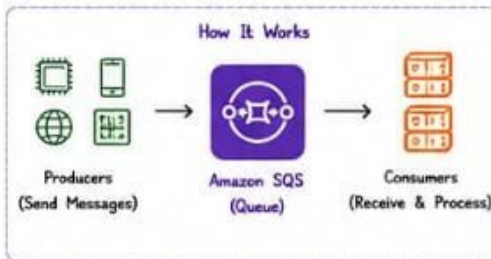


Amazon SQS (Simple Queue Service) is a fully managed messaging service that enables you to decouple and scale microservices, distributed systems, and serverless applications.

- ✓ Fully Managed
- ✓ Decoupled & Scalable
- ✓ Reliable Messaging
- ✓ High Availability
- ✓ Secure & Durable
- ✓ Cost-Effective

1. WHAT IS SQS?

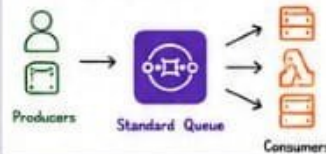
- Fully managed message queuing service.
- Decouples application components.
- Enables asynchronous communication.
- Messages are stored redundantly across multiple AZs.
- Supports standard & FIFO queues.



2. SQS QUEUE TYPES

STANDARD QUEUE

- At-least-once delivery
- Best-effort ordering
- High throughput
- Nearly unlimited TPS
- Use cases: Logging, notifications, background jobs, etc.



FIFO QUEUE

- Exactly-once processing
- First-In-First-Out ordering
- High throughput with batching
- Limited TPS (300 msg/s with batching)
- Use cases: Payments, order processing, financial transactions, etc.

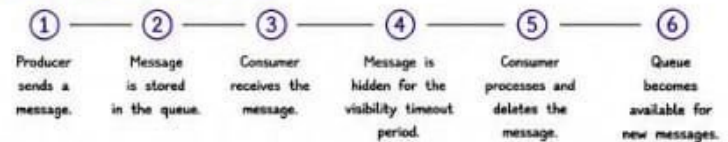


TIP: Choose Standard for high throughput & best-effort ordering.
Choose FIFO for strict ordering & exactly-once processing.

3. SQS COMPONENTS

COMPONENT	DESCRIPTION
Producer	Sends messages to the queue.
Queue	Stores messages until they are processed.
Consumer	Receives and processes messages from the queue.
Message	Unit of information sent through the queue.
Visibility Timeout	Time for which a message is hidden from other consumers.
Dead-Letter Queue (DLQ)	Stores messages that can't be processed successfully.

4. SQS MESSAGE FLOW



What happens if processing fails?

- Message is not deleted.
- After visibility timeout, it becomes visible again.
- It can be retried.
- If it exceeds the max receive count, it is moved to DLQ (if configured).



5. KEY FEATURES

Durability	Messages are stored redundantly across multiple AZs.
Scalability	Automatically scales to handle millions of messages.
Security	Encryption at rest (SSE-SQS / SSE-KMS) and in transit (HTTPS).
Access Control	IAM policies for fine-grained access control.
Delay Queues	Delay message delivery by 0 sec to 15 minutes.
Long Polling	Reduces empty responses and lowers cost.
Message Attributes	Add metadata to messages.
Dead-Letter Queues	Handle failed messages gracefully.

6. COMMON INTERVIEW QUESTIONS

- Q. What is Amazon SQS?
- A. A fully managed message queuing service that enables decoupling and scaling.
- Q. What are the types of queues in SQS?
- A. Standard and FIFO queues.
- Q. Difference between Standard and FIFO queues?
- A. Standard: best-effort ordering, at-least-once delivery, nearly unlimited TPS.
FIFO: exactly-once processing, ordered delivery, limited TPS.
- Q. What is visibility timeout?
- A. The time a message is hidden after being received by a consumer.
- Q. What is a Dead-Letter Queue (DLQ)?
- A. A queue that stores messages that cannot be processed after multiple attempts.
- Q. How does long polling work?
- A. It waits for messages to arrive in the queue before returning a response.
- Q. How do you ensure exactly-once processing?
- A. Use FIFO queue with content-based deduplication.
- Q. How do you secure SQS?
- A. Use IAM policies, encryption, VPC endpoints, and KMS keys.



7. SQS INTEGRATIONS

AWS Lambda Trigger Functions	Amazon SNS Fan-out messaging	Amazon EventBridge Event-driven architecture	Amazon S3 Upload notifications
Amazon CloudWatch Monitoring & Alarms	AWS Step Functions Orchestrate workflows	Amazon API Gateway Async request handling	AWS Fargate / ECS Decoupled workloads

9. SQS LIMITS (DEFAULT)

LIMIT	DEFAULT VALUE
Maximum Message Size	256 KB
Visibility Timeout	0 sec - 12 hours
Delay Queue	0 sec - 15 minutes
Message Retention	1 minute - 14 days
In-Flight Messages (Standard)	~120,000 (soft limit)
In-Flight Messages (FIFO)	30,000 (soft limit)
FIFO Throughput	300 msg/s (with batching)
Standard Throughput	Nearly unlimited

10. SQS PRICING (STANDARD)

- \$0.40 per 1 Million Requests
- \$0.0000004 per GB-Month of payload data



TIP: Use long polling and batching to reduce the number of requests and optimize cost.

8. BEST PRACTICES

- ✓ Use the right queue type for your use case.
- ✓ Set appropriate visibility timeout.
- ✓ Use long polling to reduce empty responses and costs.
- ✓ Implement DLQ for failed messages.
- ✓ Use message attributes to pass metadata.
- ✓ Encrypt messages using SSE-SQS or SSE-KMS.
- ✓ Monitor metrics (ApproximateNumberOfMessages, AgeOfOldestMessage, etc.).
- ✓ Use IAM roles with least privilege.
- ✓ Batch messages for better performance and lower cost.

Important Metrics

- ApproximateNumberOfMessages
- ApproximateNumberOfMessagesNotVisible
- ApproximateNumberOfMessagesDelayed
- AgeOfOldestMessage
- NumberOfMessagesSent
- NumberOfMessagesDeleted
- NumberOfEmptyReceives
- SentMessageSize / ReceivedMessageSize



KEY TAKEAWAY:

SQS helps you build decoupled, scalable, and reliable architectures.
Choose the right queue type, configure DLQ, and monitor metrics for smooth operations.

- VERIQTA

AWS MONITORING INTERVIEW GUIDE



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AWS provides powerful services to monitor, log, audit, and analyze your cloud environment. These are critical for troubleshooting, security, and operational excellence.

- ✓ Monitor Resources & Applications
- ✓ Centralized Logging & Alerts
- ✓ Audit & Compliance
- ✓ Analyze Performance & Issues
- ✓ Improve Reliability & Security

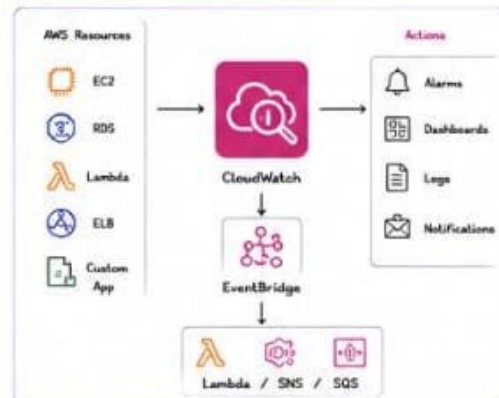
1. MONITORING SERVICES OVERVIEW

SERVICE	PURPOSE
CloudWatch	Monitoring, metrics, alarms, dashboards, logs, events.
CloudTrail	Audit API calls and account activity.
Config	Track resource changes and configuration history.
X-Ray	Analyze and troubleshoot distributed applications.

2. AMAZON CLOUDWATCH

Key Features	
Metrics	Collect metrics from AWS services and custom apps.
Alarms	Set alarms on metrics and get notified.
Logs	Collect, monitor, and store logs using Log Groups.
Dashboards	Visualize metrics and logs in custom dashboards.
Events (EventBridge)	React to changes in resources and services.

3. CLOUDWATCH ARCHITECTURE



4. CLOUDWATCH METRICS & ALARMS

Metric Types

- AWS/Service metrics (e.g., CPUUtilization, S3Error)
- Custom metrics (From applications)
- High resolution metrics (1-second granularity)

Alarm States



OK

Metric is within the threshold.



ALARM

Metric is outside the threshold.



INSUFFICIENT DATA

Not enough data to determine the state.



TIP: Use Composite Alarms to combine multiple alarms and reduce noise.

5. CLOUDWATCH LOGS

- Store and search logs from AWS services, applications, and on-premises sources.
- Use Log Groups to organize logs.
- Use Logs Insights to run queries on logs.

Log Flow



TIP: Set up log retention policies to manage costs and compliance.

6. CLOUDTRAIL (AUDIT & COMPLIANCE)

- Records API calls and account activity.
- Helps with security analysis, troubleshooting, and compliance.
- Logs delivered to S3 for long-term storage.

CloudTrail Flow



TIP: Enable CloudTrail in all regions for complete visibility.

7. AWS CONFIG (CHANGE TRACKING)

- Records resource configuration changes.
- Evaluates configurations against desired states.
- Helps with compliance and governance.

Config Components

Recorder	Records resource changes.
Configuration Item	Snapshot of a resource configuration.
Delivery Channel	Sends configuration items to S3 / CloudWatch Logs.
Rules	Evaluates configurations and checks compliance.



TIP: Use AWS Config Rules to automate compliance checks.

8. AWS X-RAY (DISTRIBUTED TRACING)

- Helps analyze and debug distributed applications.
- Visualizes requests flowing through components.
- Identifies performance bottlenecks and errors.

Tracing Flow



Key Capabilities

- ✓ Service map visualization
- ✓ Request tracing and timeline
- ✓ Error analysis and insights
- ✓ Performance bottleneck detection

9. BEST PRACTICES

- ✓ Set meaningful CloudWatch alarms and notify the right people.
- ✓ Use log retention policies to control costs.
- ✓ Centralize logs from all accounts using CloudWatch Cross-Account Observability.
- ✓ Enable CloudTrail organization trails.
- ✓ Review Config rules regularly and remediate non-compliant resources.
- ✓ Use X-Ray to trace and optimize critical user flows.
- ✓ Automate responses using EventBridge and Lambda.
- ✓ Follow least privilege for monitoring and logging services.



TIP: Monitoring is not just about setting alarms, it's about gaining actionable insights!

10. COMMON INTERVIEW QUESTIONS

- Q. What is Amazon CloudWatch?
- A. A monitoring service for metrics, logs, alarms, and dashboards.
- Q. How do you create an alarm in CloudWatch?
- A. Choose a metric, set threshold, define actions, and create the alarm.
- Q. How is CloudTrail different from CloudWatch?
- A. CloudTrail logs API calls for auditing, while CloudWatch monitors resources and applications.
- Q. What is AWS Config used for?
- A. To track resource configurations and ensure compliance.
- Q. What is AWS X-Ray?
- A. A service for tracing and analyzing requests in distributed applications.



11. COMMON METRICS (EXAMPLES)

SERVICE	COMMON METRICS
EC2	CPUUtilization, NetworkIn, NetworkOut, DiskReadOps
RDS	CPUUtilization, FreeStorageSpace, DatabaseConnections
ALB	RequestCount, HTTPCode_ELB_5XX_Count, TargetResponseTime
Lambda	Invocations, Duration, Errors, Throttles
S3	BucketSizeBytes, NumberOfObjects, 4xxErrors, 5xxErrors



TIP: Know the key metrics for each service before interviews.

12. USE CASES

- Detect and resolve performance issues before users are affected.
- Automate operational tasks and incident responses.
- Ensure security and compliance with audit trails.
- Optimize cost by monitoring usage and setting alerts.



KEY TAKEAWAY:

CloudWatch, CloudTrail, Config, and X-Ray give you deep visibility into your AWS environment. Master them to troubleshoot faster and operate with confidence.

- VERIQTA

AWS SECURITY INTERVIEW GUIDE

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- ✓ Protect Data & Infrastructure
- ✓ Manage Secrets & Encryption
- ✓ Detect Threats & Vulnerabilities
- ✓ Monitor & Respond to Security Events
- ✓ Follow Security Best Practices



AWS provides a wide range of security services and best practices to help you protect your data, workloads, and infrastructure across the cloud.

1. AWS KMS (KEY MANAGEMENT SERVICE)

FEATURE	DESCRIPTION
Create & manage encryption keys (CMKs)	Create customer managed keys for encryption.
Encrypt data	Encrypt data at rest and in transit.
Control access	Use IAM policies and key policies to control usage.
Integrations	Used by S3, EBS, RDS, Lambda, DynamoDB, and more.

TIP: Use KMS CMKs for all sensitive data and rotate keys regularly.

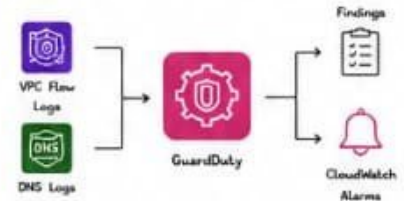
2. AWS SECRETS MANAGER

FEATURE	DESCRIPTION
Store Secrets	Store database credentials, API keys, and other secrets securely.
Automatic Rotation	Rotate secrets automatically based on schedule.
Fine-grained Access	Control access using IAM policies.
Integrations	Integrates with RDS, Redshift, DocumentDB, and more.

TIP: Use Secrets Manager instead of hardcoding credentials in applications.

3. AMAZON GUARDDUTY

- Intelligent threat detection service.
- Continuously monitors for malicious activity and unauthorized behavior.
- Uses ML, anomaly detection, and threat intelligence.
- Produces findings and severity levels.



TIP: Enable GuardDuty in all regions for continuous threat detection.

4. AMAZON INSPECTOR

- Automated security assessment service.
- Identifies vulnerabilities and deviations from best practices.
- Uses agents or agentless scans.
- Generates findings with severity and remediation.



TIP: Run regular scans and fix high severity vulnerabilities first.

5. AWS SECURITY HUB

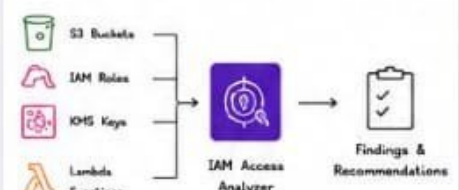
- Centralized security and compliance service.
- Aggregates findings from multiple AWS services and partner tools.
- Provides a security score and compliance status.
- Supports standards: CIS, PCI DSS, ISO 27001, etc.



TIP: Use Security Hub for single pane of glass view of your security posture.

6. IAM ACCESS ANALYZER

- Helps you identify resources shared with external entities.
- Detects public or cross-account access.
- Works with IAM policies, S3 buckets, KMS keys, etc.
- Provides actionable recommendations.



TIP: Regularly review findings and remove unwanted external access.

7. ENCRYPTION OVERVIEW

TYPE	DESCRIPTION	EXAMPLES
At Rest	Data stored in AWS is encrypted.	S3, EBS, RDS, DynamoDB (SSE-KMS)
In Transit	Data moving between services or over internet is encrypted.	HTTPS, TLS, VPN, Direct Connect
Client Side Encryption	Data encrypted before sending to AWS.	AWS KMS, SDKs, Client Libraries
Envelope Encryption	Data encrypted with data key, and data key encrypted with CMK.	KMS Envelope Encryption

TIP: Always encrypt sensitive data at rest and in transit.

8. SECURITY BEST PRACTICES

- ✓ Follow the principle of least privilege.
- ✓ Use MFA for all IAM users.
- ✓ Use IAM roles instead of access keys.
- ✓ Encrypt data at rest and in transit.
- ✓ Enable logging and monitoring.
- ✓ Regularly rotate secrets and keys.
- ✓ Keep AWS services and software updated.
- ✓ Enable multi-factor authentication (MFA).
- ✓ Use VPC, Security Groups, and NACLs to restrict access.
- ✓ Continuously monitor with GuardDuty and Security Hub.
- ✓ Perform regular security audits and reviews.

TIP: Security is a shared responsibility between AWS and you.

9. SECURITY RESPONSIBILITY MODEL

AWS (Our Responsibility)	CUSTOMER (Your Responsibility)
Security OF the Cloud	Security IN the Cloud
Global Infrastructure	Data
Hardware	Applications
Networking	Operating Systems
Virtualization	Network Configuration
Managed Services	Identity & Access Management

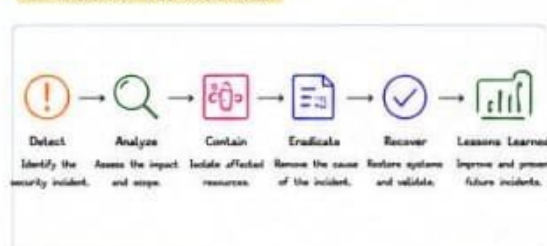
TIP: Understand the shared responsibility model for every service you use.

10. COMMON SECURITY THREATS

THREAT	DESCRIPTION
Compromised Credentials	Stolen or weak credentials lead to unauthorized access.
Exposed Resources	Publicly accessible S3 buckets, open ports, or misconfigured IAM.
Unpatched Systems	Missing updates lead to vulnerabilities.
Malicious Activity	Brute force attacks, malware, phishing, and data exfiltration.
Insider Threats	Unauthorized actions by insiders.

TIP: Prevent, Detect, and Respond to security threats.

11. INCIDENT RESPONSE FLOW



TIP: Have an incident response plan and test it regularly.

12. COMMON INTERVIEW QUESTIONS

- Q. What is AWS KMS?
- A. A managed service to create and manage encryption keys.
- Q. How does Secrets Manager help?
- A. It securely stores and rotates secrets like passwords and API keys.
- Q. What is GuardDuty?
- A. A threat detection service that monitors for malicious activity.
- Q. How is Inspector different from GuardDuty?
- A. Inspector finds vulnerabilities; GuardDuty detects threats.
- Q. What is Security Hub?
- A. It aggregates security findings and provides a security posture view.
- Q. What is IAM Access Analyzer?
- A. It identifies resources shared with external entities.



KEY TAKEAWAY:

Security is not a product, it's a process. Use the right AWS security services, follow best practices, and stay proactive to protect your cloud.

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AWS NETWORKING INTERVIEW GUIDE



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AWS Networking services help you build secure, scalable, and highly available networks in the cloud and connect your on-premise environments to AWS.

- ✓ Secure Connectivity
- ✓ Hybrid Cloud Connectivity
- ✓ Network Isolation & Routing
- ✓ High Availability
- ✓ Performance & Scalability

NETWORKING

1. TRANSIT GATEWAY

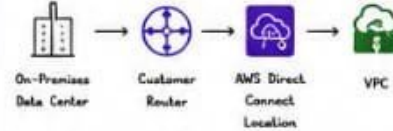
- Connects multiple VPCs and on-premises networks through a central hub.
- Simplifies network management at scale.
- Supports inter-region peering.



TIP: Use TGW when connecting 3+ VPCs or multiple on-premises networks.

2. AWS DIRECT CONNECT

- Establishes a dedicated network connection from your data center to AWS.
- Provides consistent network performance.
- More secure and reliable than internet VPN.



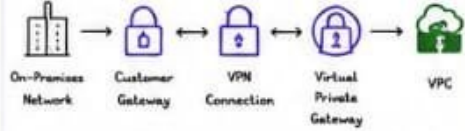
Benefits

- ✓ Higher bandwidth
- ✓ Lower latency
- ✓ More secure
- ✓ Predictable network performance

TIP: Use Direct Connect for mission-critical workloads and large data transfers.

3. VPN CONNECTION (SITE-TO-SITE)

- Creates an encrypted connection over the internet.
- Connects on-premises networks to AWS.
- Cost-effective for low to moderate bandwidth.



Types

- IPsec VPN (most common)
- SSL VPN (client VPN)
- Supports high availability with 2 VPN tunnels

TIP: Use VPN for backup to Direct Connect or for smaller environments.

4. VPC PEERING

- Connects two VPCs privately using the AWS backbone.
- No transitive routing.
- Works between accounts and regions.



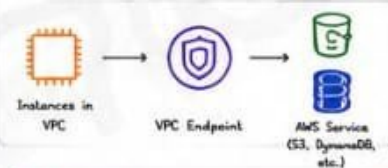
Key Points

- CIDR ranges must not overlap.
- Route tables must be updated in both VPCs.
- No bandwidth charges between peered VPCs.

TIP: Best for simple, one-to-one VPC connections.

5. VPC ENDPOINTS

- Allows private connectivity to AWS services without using the internet.
- Keeps traffic within the AWS network.
- Improves security and reduces costs.



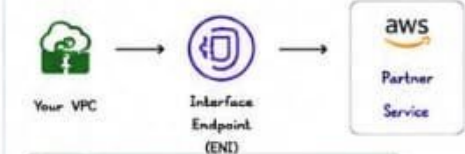
Endpoint Types

- Gateway Endpoint - S3, DynamoDB (route table based)
- Interface Endpoint - Other services (ENI based)

TIP: Use Gateway Endpoint for S3/DynamoDB and Interface Endpoint for other AWS services.

6. PRIVATELINK (AWS PRIVATE LINK)

- Provides private connectivity to supported AWS services and partner services.
- Access services without using internet, NAT, VPN, or peering.



Benefits

- Enhances security
- Simplifies network architecture
- Reduces data exposure to the internet

TIP: Use PrivateLink to access SaaS or AWS services privately and securely.

7. NETWORK ACCESS CONTROL LIST (NACL)

- Stateless firewall for controlling traffic at the subnet level.
- Rules are numbered and evaluated in order.
- Allow and Deny rules.
- Applies to both inbound and outbound traffic.

Example

Rule #	Type	Protocol	Port	Source / Destination (Subnet/CIDR)	Allow / Deny
100	Inbound	TCP	80	0.0.0.0/0	ALLOW
110	Inbound	TCP	22	203.0.113.0/24	ALLOW
120	Inbound	ALL	ALL	0.0.0.0/0	DENY
100	Outbound	ALL	ALL	0.0.0.0/0	ALLOW

TIP: Use NACL for subnet-level security. It is stateless and evaluated in order.

8. SECURITY GROUP (REVIEW)

- Stateful firewall for controlling traffic at instance level.
- Rules are evaluated before NACL.
- Allow rules only (Deny is implicit).
- Applies per ENI (instance level).

Example

Type	Protocol	Port	Source	State
Inbound	TCP	80	0.0.0.0/0	ALLOW
Inbound	TCP	22	203.0.113.0/24	ALLOW
Outbound	ALL	ALL	0.0.0.0/0	ALLOW

TIP: Use Security Groups for instance-level control. It is stateful and simpler to manage.

9. BEST PRACTICES

- ✓ Use private subnets for resources not accessible from the internet.
- ✓ Use NAT Gateway for outbound internet access from private subnets.
- ✓ Use VPC endpoints to keep traffic within AWS network.
- ✓ Use Transit Gateway for hub-and-spoke architecture.
- ✓ Use Direct Connect for high bandwidth and low latency.
- ✓ Use VPN as a backup to Direct Connect.
- ✓ Apply least privilege for network access.

TIP: Design for security, high availability, and cost efficiency.

10. COMMON INTERVIEW QUESTIONS

- Q. What is the difference between Transit Gateway and VPC Peering?
- Q. When would you use Direct Connect vs VPN?
- Q. What are the types of VPC Endpoints?
- Q. What is the difference between NACL and Security Group?
- Q. Can we have transitive routing in VPC Peering?
- Q. How does PrivateLink work?
- Q. How do you secure traffic between on-premise and AWS?
- A. What is the use of NAT Gateway?



11. SERVICE COMPARISON

SERVICE	BEST FOR	SCOPE	TYPE
Transit Gateway	Multi-VPC & Hybrid Hub	VPC / On-Prem	Hub
Direct Connect	High BW, Low Latency	On-Prem ↔ AWS	Dedicated
VPN	Cost-effective connectivity	On-Prem ↔ AWS	Encrypted
VPC Peering	Simple VPC to VPC	VPC ↔ VPC	1:1
VPC Endpoint	Private access to AWS services	VPC ↔ AWS APIs	Private
PrivateLink	Private access to AWS/Partner	VPC ↔ Service	Private

12. QUICK REFERENCE

Icon	Service	Description
TGW	Transit Gateway	Central hub for VPCs & on-prem
DC	Direct Connect	Dedicated private connection
VPN	VPN	Encrypted connection over internet
Peering	VPC Peering	VPC to VPC private connection
Endpoint	VPC Endpoint	Private access to AWS services
PrivateLink	PrivateLink	Private access to AWS/partner
NACL	NACL	Subnet-level, stateless firewall
SG	Security Group	Instance-level, stateful firewall



KEY TAKEAWAY:

AWS Networking services help you connect, secure, and isolate your resources efficiently. Choose the right service based on security, performance, and scalability needs.

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AWS HIGH AVAILABILITY & DISASTER RECOVERY INTERVIEW GUIDE



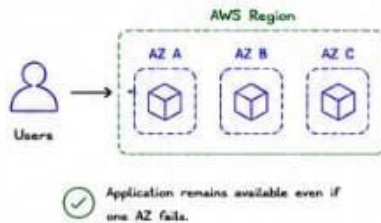
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- ✓ High Availability by Design
- ✓ Disaster Recovery Strategies
- ✓ Multi-AZ & Multi-Region
- ✓ RTO, RPO & Recovery Strategies
- ✓ Backup, Restore & Resilience

★ AWS provides multiple services and design patterns to help you build highly available systems and recover quickly from failures or disasters with minimal data loss.

1. HIGH AVAILABILITY (HA) OVERVIEW

- Ensures your application is always available.
- No single point of failure.
- Uses redundancy across Availability Zones (AZs).
- Improves fault tolerance and uptime.



TIP: Design for failure. Assume something will fail.

2. DISASTER RECOVERY (DR) OVERVIEW

- Recovers IT systems after a major disruption.
- Focuses on data protection and business continuity.
- Involves backup, replication, and failover.
- Measured by RTO and RPO.

TERM	MEANING
RTO (Recovery Time Objective)	How quickly a system must be restored after a disaster. (e.g., 15 minutes)
RPO (Recovery Point Objective)	How much data loss is acceptable. (e.g., 5 minutes of data)

TIP: Lower RTO and RPO usually mean higher cost.

3. AWS AVAILABILITY CONCEPTS

Region	A physical location in the world (e.g., us-east-1).
Availability Zone (AZ)	Isolated data center within a region with independent power, network, and cooling.
Edge Location	Used by CloudFront for caching content closer to users.
Local Zone	Extends AWS services closer to population centers.
Wavelength Zone	Deploys workloads on 5G edge networks.

TIP: Use multiple AZs in a Region for HA. Use multiple Regions for DR.

4. HA ARCHITECTURE PATTERNS

Multi-AZ	Deploy resources across two or more AZs in the same Region.
Auto Scaling	Automatically replace unhealthy instances across AZs.
Load Balancing	Distributes traffic across healthy instances in multiple AZs.
Database Multi-AZ	Use RDS Multi-AZ or Aurora Multi-AZ for automatic failover.
ElasticCache Cluster Mode	Distribute cache nodes across AZs.
S3 Cross-Region Replication (CRR)	Replicate data between Regions for durability.

TIP: Combine these patterns for a resilient architecture.

5. COMMON AWS SERVICES FOR HA

SERVICE	HOW IT HELPS
ELB / ALB / NLB	Distributes traffic across AZs.
Auto Scaling	Maintains desired capacity.
RDS Multi-AZ	Automatic failover to standby.
Aurora Multi-AZ	Storage replicated 6 ways.
EFS	Regional service with multi-AZ storage.
DynamoDB	Replicates across 3 AZs by default.
S3	11 9s durability across AZs.
Route 53	Health checks + failover routing.

TIP: Choose managed services that handle redundancy for you.

6. DISASTER RECOVERY STRATEGIES

Backup & Restore	Restore from backup when disaster occurs. (Lowest cost, higher RTO)
Pilot Light	Minimal resources running. Scale up during disaster.
Warm Standby	Scaled-down version running. Faster recovery.
Multi-Site (Active-Passive)	Standby site ready to take over.
Multi-Region (Active-Active)	Traffic served from multiple Regions. Highest availability.

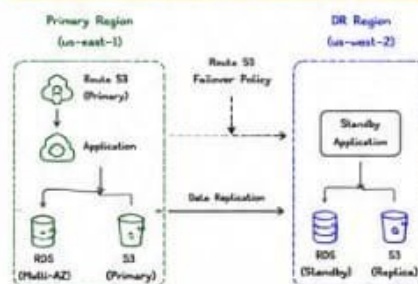
TIP: Choose strategy based on business needs, RTO, RPO, and budget.

7. MULTI-AZ vs MULTI-REGION

	MULTI-AZ	MULTI-REGION
Scope	Within a Region	Across Regions
Use Case	HA, Fault Tolerance	DR, Global Applications
Latency	Low	Higher
Cost	Lower	Higher
RPO	Very low	Configurable
Example	RDS Multi-AZ	Active-Passive DR

TIP: Multi-AZ protects from AZ failure. Multi-Region protects from regional disasters.

8. DR ARCHITECTURE EXAMPLE (ACTIVE-PASSIVE)



TIP: Use Route 53 Failover routing and data replication for DR.

9. BACKUP & RECOVERY OPTIONS

SERVICE	BACKUP OPTIONS
EC2	EBS Snapshots, AMIs
RDS / Aurora	Automated Backups, Manual Snapshots, Cross-Region Read Replicas
DynamoDB	Point-in-Time Recovery (PITR), On-Demand Backups
S3	Versioning, Cross-Region Replication, Lifecycle Policies
EFS	AWS Backup, EFS-to-EFS Replication

TIP: Test your backups and recovery process regularly!

10. BEST PRACTICES

- ✓ Design for failure, not for success.
- ✓ Use multiple AZs for HA.
- ✓ Use Multi-Region for DR.
- ✓ Automate backups and replication.
- ✓ Test DR plan regularly.
- ✓ Monitor health and set up alerts.
- ✓ Document your DR runbook.
- ✓ Keep your infrastructure as code.

TIP: A tested DR plan is better than a perfect plan that was never tested.

11. COMMON INTERVIEW QUESTIONS

- Q: What is the difference between HA and DR?
A: HA prevents downtime in the same Region. DR recovers from a complete regional outage.
- Q: What is RTO and RPO?
A: RTO is how fast you recover. RPO is how much data you can lose.
- Q: How would you achieve HA for a web application?
A: Use ALB, Auto Scaling, and Multi-AZ RDS.
- Q: How would you implement DR for critical data?
A: Use backup, replication, and Route 53 Failover with a standby Region.
- Q: What strategy would you use for a banking application?
A: Multi-Region (Active-Passive) with low RTO and RPO.



12. KEY METRICS TO REMEMBER

- Availability = Uptime / Total Time
- RTO = Recovery Time Objective
- RPO = Recovery Point Objective
- 99.99% Uptime = ~52 minutes downtime/year
- 99.999% Uptime = ~5 minutes downtime/year

★ KEY TAKEAWAY:

Build for failure, automate recovery, and test your plan. AWS gives you the building blocks to achieve high availability and recover quickly from any disaster.

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AWS ARCHITECTURE INTERVIEW SCENARIOS

Design Scalable, Highly Available, Cost-Effective Solutions



Interviewers love architecture questions. They test your ability to design, choose the right AWS services, and justify your decisions.

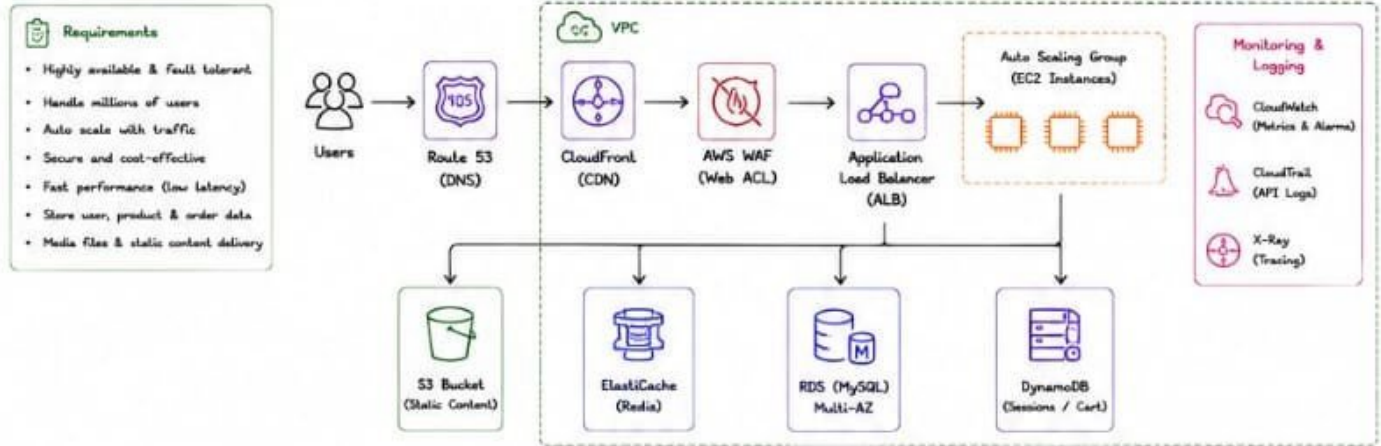


ARCHITECTURE

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- ✓ Understand Requirements
- ✓ Design Scalable Architecture
- ✓ Choose Right AWS Services
- ✓ Explain Trade-offs
- ✓ Justify & Optimize

1. SCENARIO: DESIGN A SCALABLE E-COMMERCE APPLICATION ON AWS



2. WHY THESE SERVICES?

SERVICE	WHY WE CHOSE IT
Route 53	Highly available DNS service to route users to the application.
CloudFront	Delivers content globally with low latency and high transfer speeds.
AWS WAF	Protects against common web exploits and DDoS attacks.
ALB	Distributes traffic across multiple targets and handles SSL termination.
Auto Scaling	Automatically scales EC2 instances based on demand.
EC2	Flexible compute instances to run application servers.
ElastiCache (Redis)	Caches frequently accessed data for high performance.
RDS Multi-AZ	Provides automated backups, failover and high availability.
DynamoDB	Stores session data and shopping cart with single-digit ms latency.
S3	Stores static files like images, CSS, JS with high durability.
CloudWatch	Monitors resources and application performance.
CloudTrail	Logs API calls for security and compliance.
X-Ray	Helps debug and trace requests across services.

3. ARCHITECTURAL PRINCIPLES APPLIED

High Availability	Multi-AZ RDS, Auto Scaling, ALB, CloudFront.
Scalability	Auto Scaling for compute, DynamoDB for high write throughput.
Performance	CloudFront, CDN, ElastiCache for caching.
Security	WAF, IAM, Security Groups, TLS, CloudTrail.
Cost Optimization	S3 for storage, Auto Scaling, CloudWatch for right-sized monitoring.

4. TRADE-OFFS TO DISCUSS

- RDS Multi-AZ increases availability but adds cost.
- CloudFront improves performance but has caching considerations.
- DynamoDB offers high performance but can be costlier at scale.
- ALB is layer 7 (HTTP/HTTPS) - not suitable for TCP level routing.

5. FOLLOW-UP QUESTIONS INTERVIEWERS MAY ASK

- How will you handle spike in traffic during sales?
- How will you ensure data durability?
- How will you handle user sessions?
- What if a single AZ goes down?
- How will you reduce latency for global users?
- How will you secure the application?
- How will you monitor and alert failures?
- How will you deploy updates without downtime?



6. BEST PRACTICE APPROACH

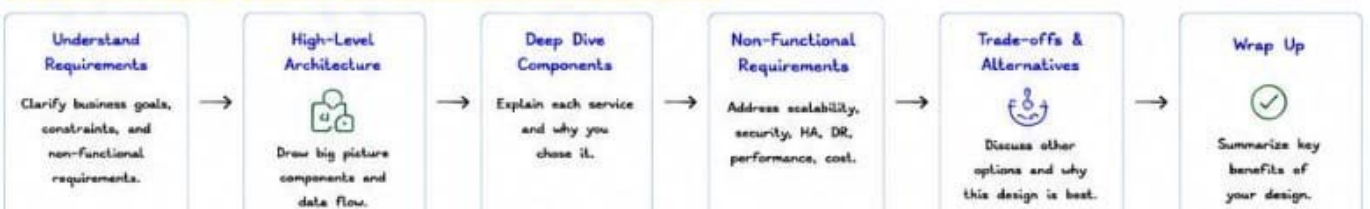
- ✓ Design for failure, not for success.
- ✓ Use managed services wherever possible.
- ✓ Follow least privilege security model.
- ✓ Automate infrastructure with IaC.
- ✓ Implement CI/CD for faster deployments.
- ✓ Monitor everything and set up alerts.
- ✓ Test your architecture for scalability.
- ✓ Document your architecture clearly.



7. OTHER COMMON SCENARIOS

- Design a real-time chat application.
- Design a video streaming platform.
- Design a banking application.
- Design a serverless API.
- Design a data analytics platform.

8. ARCHITECTURE ANSWER TEMPLATE (USE THIS STRUCTURE IN INTERVIEWS)



KEY TAKEAWAY:

Great architects understand requirements, design for reliability, and justify every choice. Practice, explain clearly, and think like an AWS Solutions Architect!

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AWS TOP 30 INTERVIEW QUESTIONS

Most Frequently Asked AWS Interview Questions with Key Points



These questions cover core AWS services, architecture, security, networking, monitoring, and best practices commonly asked in AWS interviews.

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BE PREPARED.
BE CONFIDENT.

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- ✓ Core Concepts
- ✓ Architecture & Design
- ✓ Security & Compliance
- ✓ Operations & Best Practices
- ✓ Real Interview Questions

1. COMPUTE & STORAGE

	Q1. What is the difference between EC2, ECS, and Lambda?	EC2 = virtual servers, full control. ECS = container orchestration. Lambda = serverless, no servers to manage.
	Q2. When would you choose EC2 over Lambda?	When you need longer running tasks, full control, specific OS, or persistent connections.
	Q3. What is the difference between EBS and Instance Store?	EBS = persistent, survives stop/start. Instance Store = temporary, lost on stop/terminate.
	Q4. What is the difference between S3 Standard and S3 Glacier?	S3 Standard = frequent access. Glacier = archival storage, very low cost, slower retrieval.
	Q5. What is EFS?	A managed NFS file system that can be shared across multiple EC2 instances.

2. NETWORKING

	Q6. What is a VPC?	A Virtual Private Cloud is a logically isolated section of AWS cloud where you can launch resources in a virtual network.
	Q7. What is the difference between Public and Private Subnet?	Public subnet has a route to an Internet Gateway. Private subnet does not.
	Q8. What is a NAT Gateway used for?	Allows instances in a private subnet to access the internet for updates, but blocks inbound traffic.
	Q9. What is the difference between ALB, NLB, and CLB?	ALB = Layer 7 (HTTP/HTTPS). NLB = Layer 4 (TCP/UDP). CLB = Classic Load Balancer (legacy).
	Q10. What is CloudFront?	A CDN service that delivers content with low latency and high transfer speeds.

3. DATABASE & ANALYTICS

	Q11. What is the difference between RDS and Aurora?	Aurora is AWS's enhanced database that is compatible with MySQL/PostgreSQL, more scalable and highly available.
	Q12. What is Multi-AZ in RDS?	RDS replicates data to a standby instance in another AZ for high availability.
	Q13. What is DynamoDB?	A fully managed NoSQL database that provides single-digit millisecond latency at any scale.
	Q14. What is Redshift?	A data warehouse service used for analytics and business intelligence.
	Q15. What is the difference between OLTP and OLAP?	OLTP = transactional systems (RDS). OLAP = analytical systems (Redshift).

4. SECURITY & IDENTITY

	Q16. What is IAM?	IAM (Identity and Access Management) helps you securely control access to AWS resources.
	Q17. What is the difference between IAM User, Role, and Policy?	User = identity with long-term credentials. Role = identity that you can assume. Policy = JSON document that defines permissions.
	Q18. What is AWS KMS?	Key Management Service used to create and manage encryption keys.
	Q19. What is AWS Secrets Manager?	Stores and manages secrets like passwords, API keys, and database credentials.
	Q20. What is GuardDuty?	A threat detection service that continuously monitors for malicious activity.

5. MONITORING & MANAGEMENT

	Q21. What is CloudWatch used for?	Monitoring metrics, logs, creating alarms, and visualizing performance.
	Q22. What is CloudTrail?	Records API calls and account activity for audit and compliance.
	Q23. What is AWS Config?	Tracks resource configuration changes and compliance over time.
	Q24. What is X-Ray?	Used to analyze and debug distributed applications.
	Q25. What is Systems Manager?	Centrally view and manage your AWS resources at scale.

6. ARCHITECTURE & BEST PRACTICES

	Q26. How do you design for high availability on AWS?	Use Multi-AZ, Auto Scaling, Load Balancers, health checks, and remove single points of failure.
	Q27. How do you design for scalability?	Use Auto Scaling, managed databases, serverless services, and decouple components.
	Q28. What is the Well-Architected Framework?	A framework with 6 pillars: Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, Sustainability.
	Q29. How do you reduce costs on AWS?	Right-size resources, use Savings Plans, S3 lifecycle policies, auto scaling, and turn off unused resources.
	Q30. How do you ensure data security in AWS?	Use IAM least privilege, encryption (KMS), security groups, VPC isolation, and monitoring services.



INTERVIEW TIPS

- ✓ Always explain your thought process before jumping into solutions.
- ✓ Use AWS Well-Architected Framework to justify your decisions.
- ✓ Give real-world examples when possible.
- ✓ Discuss trade-offs: cost, performance, security, and complexity.
- ✓ If you don't know, be honest and explain how you would find the answer.

HOW TO ANSWER EFFECTIVELY

- ✓ Clarify the question and requirements.
- ✓ Break down the problem into components.
- ✓ Explain your approach and trade-offs.
- ✓ Summarize the solution and benefits.



REMEMBER

- ✓ Think like an architect.
- ✓ Be clear, structured and confident.
- ✓ Security, Scalability, Reliability.
- ✓ Cost Optimization always matters.



KEY TAKEAWAY:

Practice these questions, design solutions, and explain your reasoning.
The more you practice, the more confident you will be in your interviews!

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AWS TROUBLESHOOTING INTERVIEW GUIDE

Common AWS Issues, Root Causes & How to Fix Them



TROUBLESHOOT LIKE A PRO

PAGE 19 OF 20

- ✓ Identify the Issue
- ✓ Analyze the Cause
- ✓ Use the Right Tools
- ✓ Apply the Fix
- ✓ Prevent Recurrence



Interviewers love scenario-based questions.

Show that you can troubleshoot systematically and use the right AWS tools to find and fix issues.

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1. EC2 INSTANCE NOT CONNECTING (SSH/RDP)

Symptoms	Common Causes	✗ How to Troubleshoot & Fix
Cannot SSH to EC2 instance.	- Security Group blocking port 22/3389 - NACL blocking traffic - Wrong key pair or password - Instance not running - Public IP not associated	- Check instance state (running) - Verify Security Group inbound rules - Check NACL inbound/outbound rules - Ensure correct key pair / password - Confirm instance has a Public IP - Try connecting from allowed IP



2. RDS DATABASE CONNECTION FAILED

Symptoms	Common Causes	✗ How to Troubleshoot & Fix
Cannot connect to RDS instance.	- Security Group not allowing access - DB not publicly accessible - Wrong endpoint/port/database - DB instance stopped - IAM authentication failure	- Check Security Group inbound rules - Verify DB Subnet Group & routing - Confirm endpoint, port, DB name - Ensure DB instance is available - Check credentials / IAM permissions



3. S3 ACCESS DENIED

Symptoms	Common Causes	✗ How to Troubleshoot & Fix
AccessDenied when accessing S3 bucket.	- IAM user/role lacks permissions - Bucket policy denies access - Block Public Access enabled - Wrong region or endpoint - Request not signed properly	- Check IAM policy permissions - Review bucket policy - Check Block Public Access settings - Verify bucket region - Use AWS CLI with correct profile



4. LOAD BALANCER UNHEALTHY TARGETS

Symptoms	Common Causes	✗ How to Troubleshoot & Fix
Targets show Unhealthy in ALB / NLB.	- Health check failing - Application not responding - Wrong target group port - Security Group blocking traffic - Target instance stopped	- Check target health in console - Verify health check path/port - Test from instance (curl/netcat) - Check Security Group rules - Ensure instances are running



5. CLOUDWATCH ALARMS NOT TRIGGERING

Symptoms	Common Causes	✗ How to Troubleshoot & Fix
Alarm not firing even when metric is high.	- Wrong metric or namespace - Wrong threshold - Period / statistics mismatch - Insufficient data points - Alarm in INSUFFICIENT_DATA state	- Verify metric and namespace - Check threshold and comparison - Review period, evaluation periods - Ensure enough data is being sent - Check alarm state & history



6. LAMBDA FUNCTION TIMING OUT

Symptoms	Common Causes	✗ How to Troubleshoot & Fix
Lambda Function timeout error.	- Function takes too long - External service slow - Memory too low - Infinite loop / retry issue - VPC networking issues	- Check CloudWatch Logs - Increase timeout value - Increase memory allocation - Optimize code / dependencies - Check VPC & NAT Gateway



7. ROUTE 53 DNS NOT RESOLVING

Symptoms	Common Causes	✗ How to Troubleshoot & Fix
Domain not resolving or wrong IP.	- Wrong record type/value - TTL caching - Hosted zone not associated - Propagation delay - Registrar nameservers incorrect	- Check record type & value - Use "dig" or "nslookup" - Verify hosted zone & NS records - Wait for propagation (TTL) - Ensure correct nameservers



8. CLOUDWATCH LOGS NOT AVAILABLE

Symptoms	Common Causes	✗ How to Troubleshoot & Fix
Logs not showing in CloudWatch.	- Logs not enabled - Wrong log group name - Insufficient IAM permissions - Retention policy expired - Application not logging	- Enable logging for the service - Check log group & stream name - Verify IAM role permissions - Check retention policy - Ensure application is logging



9. VPC NO INTERNET ACCESS

Symptoms	Common Causes	✗ How to Troubleshoot & Fix
EC2 instance cannot access the internet.	- No Internet Gateway - Route table missing default route - NACL blocking outbound - Security Group blocking traffic - No NAT Gateway (private subnet)	- Attach Internet Gateway - Add 0.0.0.0/0 route to IGW - Check NACL outbound rules - Check Security Group outbound - For private subnet, use NAT GW



10. KMS ACCESS DENIED

Symptoms	Common Causes	✗ How to Troubleshoot & Fix
AccessDenied when using KMS key.	- IAM policy lacks kms permissions - Key policy denies access - Wrong key or region - Cross-account permission issue - Disabled key	- Check IAM policy for KMS actions - Review KMS key policy - Verify key ID and region - Grant cross-account permissions - Enable the key



11. ESSENTIAL AWS TOOLS FOR TROUBLESHOOTING

CloudWatch Monitor metrics, logs, alarms and gain visibility.	CloudTrail Track API calls and user activity.	VPC Flow Logs Capture network traffic for analysis.	AWS Config Check resource configuration changes.	IAM Access Analyzer Identify access issues and overly permissive policies.	AWS X-Ray Trace requests and identify bottlenecks.
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GENERAL TROUBLESHOOTING APPROACH



PRO TIPS

- ✓ Always check CloudWatch Logs first.
- ✓ Reproduce the issue in a safe environment.
- ✓ Use the AWS Well-Architected Framework.
- ✓ Automate monitoring and alerting.
- ✓ Document everything you learn!



KEY TAKEAWAY:

Great engineers don't just build, they troubleshoot and resolve issues fast. Stay calm, follow the process, use the right tools, and fix the root cause!

— VERIQTA —

QUICK REFERENCE FOR CONFIDENT INTERVIEWS



A one-page (well, almost!) reference of high-value facts, limits, ports, models, and best practices to review before any AWS interview.

— VERIQTA —



CHEAT SHEET

- ✓ High-Value Quick Reference
- ✓ Service Limits & Quotas
- ✓ Ports & Protocols
- ✓ Well-Architected Reminders
- ✓ Last-Minute Revision Guide

1. AWS GLOBAL INFRASTRUCTURE

Regions	A physical location in the world (e.g., us-east-1).
Availability Zones (AZs)	Isolated data centers within a Region with independent power, network, and cooling.
Edge Locations	Used by CloudFront for caching content closer to users.
Local Zones	Extend AWS services closer to population centers.
Wavelength Zones	For ultra low latency 5G applications.



TIP: Choose the right Region & AZs for high availability and low latency.

2. COMMON AWS SERVICE CATEGORIES

Compute	EC2, Lambda, ECS, EKS, Elastic Beanstalk, Lightsail
Storage	S3, EBS, EFS, Glacier, Storage Gateway
Database	RDS, Aurora, DynamoDB, ElastiCache, Redshift
Networking & Content Delivery	VPC, Route 53, CloudFront, API Gateway, ELB
Security, Identity & Compliance	IAM, KMS, Secrets Manager, GuardDuty, Security Hub
Monitoring & Management	CloudWatch, CloudTrail, AWS Config, Systems Manager



TIP: Understand which category a service belongs to and how it fits in the architecture.

3. COMMON PORTS & PROTOCOLS

PORT	PROTOCOL	SERVICE / USE
22	TCP	SSH (EC2 Linux)
21	TCP	FTP
80	TCP	HTTP (Web)
443	TCP	HTTPS (Secure Web)
3306	TCP	MySQL / Aurora MySQL
5432	TCP	PostgreSQL / Aurora PostgreSQL
1433	TCP	Microsoft SQL Server
27017	TCP	MongoDB
3389	TCP	RDP (Windows)
1024-65535	TCP/UDP	Ephemeral Ports (Outgoing)



TIP: Security Groups are stateful. NACLs are stateless. Allow the minimum ports required.

4. AWS RESPONSIBILITY MODEL

AWS (SECURITY OF THE CLOUD)	YOU (SECURITY IN THE CLOUD)
Global Infrastructure	Your Data
Regions, AZs, Edge Locations	Identity & Access Management
Physical Security	Operating Systems
Hardware, Software, Networking	Network Configuration
Virtualization Layer	Security Groups & NACLs
Managed Services	Data Encryption at Rest
Platform Services	Application & Content



TIP: You secure what is in the cloud. AWS secures the cloud.

5. AWS PRICING MODELS

On-Demand	Pay for what you use by the hour/second.
Reserved Instances	Commit 1 or 3 years. Lower cost.
Savings Plans	Commit for compute usage (EC2, Fargate, Lambda).
Spot Instances	Unused capacity at 90%+ discount.
Dedicated Hosts	Physical server for your use.



TIP: Use the right pricing model to reduce costs significantly.

6. HIGH AVAILABILITY QUICK GUIDE

Multi-AZ	Deploy across multiple AZs in the same Region.
Auto Scaling	Automatically adjust capacity based on demand.
Load Balancing	Distribute traffic across healthy instances.
Data Replication	Use RDS Multi-AZ, DynamoDB Global Tables, S3 CRR.
Route 53 Health Checks	Automatically reroute traffic on failure.



TIP: Design for failure, not just for success.

7. WELL-ARCHITECTED FRAMEWORK (6 PILLS)

Operational Excellence	Run and monitor systems effectively and improve continuously.
Security	Protect data, systems, and assets.
Reliability	Maintain workload performance and recover from failure.
Performance Efficiency	Use resources efficiently to meet performance needs.
Cost Optimization	Avoid unnecessary costs and optimize over time.
Sustainability	Minimize environmental impact.



TIP: Use these pillars to answer architecture questions.

8. SERVICE LIMITS (DEFAULT EXAMPLES)

SERVICE	DEFAULT LIMIT (PER REGION)
EC2 Running Instances	~20 On-Demand instances
EBS Volumes per Instance	~39
ELB (ALB) per Region	~50
Lambda Concurrent Executions	~1,000
S3 Buckets	~100
DynamoDB Tables	~40
VPCs	~5
Security Groups (per VPC)	~2,500
Route 53 Hosted Zones	~25



TIP: You can request quota increases in the Service Quotas console.

9. COMMON DECISION GUIDE

NEED	USE
Virtual Servers	EC2
Run Code Without Servers	Lambda
Store Files / Objects	S3
Block Storage for EC2	EBS
File Storage (Shared)	DFS
Relational Database	RDS / Aurora
NoSQL Database	DynamoDB
Cache for Apps	ElastiCache
DNS Management	Route 53
API Management	API Gateway
Message Queuing	SQS / SNS



TIP: Understand the use case first, then choose the service.

10. BEST PRACTICES (QUICK REMINDER)

- ✓ Use least privilege IAM policies.
- ✓ Enable MFA for privileged accounts.
- ✓ Encrypt data at rest and in transit.
- ✓ Use Multi-AZ for critical workloads.
- ✓ Enable CloudWatch Alarms.
- ✓ Back up your data regularly.
- ✓ Keep resources tagged.
- ✓ Review costs and optimize.
- ✓ Automate with Infrastructure as Code.
- ✓ Test your DR and backup plans.

11. LAST-MINUTE INTERVIEW CHECKLIST

- ✓ Revise core services and their use cases.
- ✓ Practice whiteboard architecture designs.
- ✓ Review security, HA, and DR concepts.
- ✓ Understand trade-offs and pricing models.
- ✓ Go through the Top 30 AWS interview questions.
- ✓ Be ready to explain your past projects.
- ✓ Stay calm, think out loud, and communicate clearly!



KEY TAKEAWAY:

Understand the fundamentals.
Design for reliability, security, and cost-effectiveness.
Keep learning. Keep building.
You've got this!



— VERIQTA —



FINAL WORD:

The more you understand AWS, the easier the interview becomes.
Focus on concepts, practice consistently, and build with confidence!

— VERIQTA —